

Astana IT University

DIPLOMA PROJECT

Pre-defense 1

Development of an AR/VR Application for Professional Training of Medical Specialists

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Project Proposal: Development of an AR/VR Application for Professional Training of Medical Specialists

Project Title

Development of an AR/VR Application for Professional Training of Medical Specialists

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1. Project proposal

1.1 Problem statement

Practical training of medical specialists depends on access to mannequins, simulation centers, instructors, and real clinical cases. High-fidelity simulation requires considerable financial investment, dedicated facilities and trained personnel [1]; mannequin-based training needs small groups and more instructors [11], and large cohorts (350–500 students) make small-group follow-up difficult to sustain [10]. As a result, opportunities for repeated practice are often limited [1].

Feedback is frequently subjective and delivered late. Individual assessment with one instructor per student is not realistic in terms of cost and resources [11], and in CPR training human assessment agreed poorly with software-based measurement [6]. In a randomized trial of VR venipuncture training, shorter feedback delay was the mechanism that mediated skill transfer [5].

AR/VR technologies make it possible to practise procedures repeatedly in a safe, controlled environment with immediate feedback [1] and to record objective performance data such as time, attempts and tool pressure [3], [13]. The evidence for learning effectiveness is, however, of low certainty, and VR is regarded as a complement to hands-on training rather than a replacement [1]. The reviewed solutions mostly cover a single procedure on a specific device or setup [5], [11], [12], [13], and none describes a common, documented format for adding new procedures (Section 3.4). This suggests that institutions needing several procedures may have to adopt several unrelated products (an assumption to be confirmed in interviews, Section 2). A common, extensible platform with a unified way to define scenarios, evaluate performance, and track progress is therefore proposed as a design approach, and this project evaluates its feasibility.

1.2 Target users

- Trainees: medical students and interns who practise procedures and receive

feedback.

- Instructors: teachers and clinical trainers who assign scenarios, review results, and monitor progress.
- Administrators: staff of an educational institution or simulation center who manage users, scenarios, and devices.

Instructor and administrator functions are planned for the next stage of the project.

1.3 Aim

To design and develop a prototype of an extensible AR/VR platform for training medical specialists in clinical procedures, with automated performance assessment, and to evaluate its feasibility using

cardiopulmonary resuscitation (CPR) as the first implemented procedure and blood draw as the second.

1.4 Objectives

1. Analyse user needs and the current state of AR/VR training in medical education (literature, existing systems, discussions with user representatives).
2. Specify functional and non-functional requirements with priorities and acceptance criteria.
3. Select key technologies by comparing alternatives against the requirements.
4. Design the platform architecture, including a procedure module that allows new scenarios to be added without changing the core system.
5. Design the user interface and key user scenarios for trainees, with instructor-related scenarios planned for the next stage.
6. Implement a working prototype with two complete procedures (CPR and blood draw), including step guidance, real-time feedback, and a results summary.
7. Implement collection of performance data (e.g. timing, correctness of steps) for assessment; persistent storage and progress tracking are planned for the next stage.
8. Define and carry out a verification and evaluation plan with measurable metrics and a baseline for comparison (reference values from clinical guidelines and expert assessment of the same sessions), including a check that the second procedure was added through the procedure module without changes to the core system.

1.5 Expected outcomes

- A requirements specification with verifiable acceptance criteria.
- An architecture description with context, use case, and component diagrams.
- UX mockups of the key scenarios.
- A working prototype of the platform with two complete procedures (CPR and blood draw).
- A performance assessment mechanism that records and presents objective results for each session.
- A verification report showing to what extent the requirements and the project aim have been achieved, including evidence that the platform can be extended with a new procedure without changing its core.

1.6 Project scope

In scope (current stage, prototype):

- Core platform: procedure selection, scenario execution, and results review for a session.
- Procedure module with a defined structure for scenarios (steps, conditions, metrics, scoring rules).
- Two fully implemented procedures: CPR and blood draw.

- Collection of performance data during a session.
- Verification of the prototype against the defined requirements and

metrics. Planned for the next stage (pre-defense 2):

- User roles (trainee, instructor, administrator).
- Persistent storage of results and progress tracking.
- Instructor view for reviewing trainee

results. Out of scope (potential future

work):

- Additional procedures beyond CPR and blood draw (e.g. patient examination, surgical procedures).
- Clinical validation with real patients or certification as a medical training product.

- Support for all AR/VR devices on the market; the prototype targets a limited set of selected devices.
- Integration with external learning management systems.

2. Evidence of Relevance

2.1 Method

- Format: structured written interviews through an online form, 19 open questions in four blocks (background, current training practice, difficulties and barriers, expectations of a VR platform). The full question list is in Appendix A.
- Period: 23-24 October 2026.
- Respondents: 3 user representatives (Table 2.1). All three gave explicit consent to use their answers; answers are anonymised.
- Purpose: to check whether the problem in the project proposal exists in practice, to collect user requirements, and to separate verified findings from assumptions.

ID	Role	Specialty / year	Experience with VR, AR or computer simulators
R1	Medical instructor	Neurology	None
R2	Medical student	5th year	None
R3	Medical student	Surgery, 4th year	None

Table 2.1. Respondents

2.2 Findings

Topic	R1 (instructor)	R2 (student, 5th year)	R3 (student, 4th year)
CPR training tools	Face-to-face training, AED, manikin	Simulator	CPR manikin
Blood collection / IV training	Trainer to trainee	Demonstration on a patient	Theory, simulation and manikin practice
Repetitions per procedure	2-3	3	"Hundreds of times" (cumulative, as stated)
Assessment	Success rate	Theory and practical exams	Timed practical check-offs; zero-tolerance checklist for safety and critical errors; live-patient evaluation
Feedback timing	Immediately	After the examination	Immediately in the lab; within minutes in clinical rotations
Hardest part	Short training time	Intense training	Performance anxiety and the tactile "feel" of a vein
Typical mistakes	Chest compression, uncertain actions	Tactical mistakes	Mechanical errors caused by nervousness or procedural shortcuts
Barriers to more practice	Fear of failing	Time limitation	Institutional cost, safety regulations, shortage of clinical placements
Useful platform features	Simulation for repeated training	Training at own pace	No answer ("I don't know")
Performance data wanted	Analysis of mistakes and speed	Mistakes and how to improve them	No answer

Barriers to using a platform	Bugs, inconvenience of wearing a headset	Availability	None stated
Motion sickness with VR	No	No	No answer

Table 2.2. Summary of answers

2.3 What the evidence supports

#	Finding	Evidence	Effect on requirements
E1	Current CPR and blood draw training is done with manikins, simulators and trainer demonstration	R1, R2, R3	Confirms the baseline for evaluation (Section 9): manikin training with instructor assessment
E2	Practice time is limited, and students want to repeat procedures	R1 (2-3 repetitions, short training time, wants repeated simulation), R2 (3 repetitions, time limitation, wants own pace), R3 (cost, safety rules and placement shortage limit practice)	Supports repeatable practice at the learner's own pace (FR-12)
E3	Assessment is checklist and time based, with critical-error rules	R1 (success rate), R3 (timed check-offs, zero-tolerance checklist)	Supports step-level scoring and time measurement (FR-8, FR-9)
E4	Feedback timing varies; one student receives it only after the examination	R2; R1 and R3 report immediate feedback in the lab	Supports immediate feedback after each step (FR-4)
E5	Typical mistakes are chest compression errors, uncertain actions, tactical mistakes and nervous mechanical errors	R1, R2, R3	Supports step order and condition checking (FR-3) and CPR compression measurement (FR-5, FR-6)
E6	Fear of failing and performance anxiety slow learning	R1, R3	Supports safe repeated practice in a virtual environment (FR-12)
E7	Users want to see mistakes, speed and how to improve	R1, R2	Supports the results summary with errors, time and score (FR-9, FR-10)
E8	Concerns: bugs, discomfort of wearing a headset, availability	R1, R2	Supports stable operation (NFR-7), comfort and session length (NFR-4, NFR-9) and a standalone headset that does not need a PC (NFR-2)
E9	The instructor wants performance data (mistakes and speed)	R1	Partial support for the planned instructor view (FR-14, FR-15, P2); not confirmed by other respondents

Table 2.3. Findings and their effect on requirements

2.4 Verified findings and assumptions

Statement	Level	Basis
Manikin-based training is the current practice	Verified (3 of 3)	E1
Practice time is limited and repeated practice is wanted	Verified (3 of 3)	E2
Feedback should be immediate and show mistakes	Verified in part (2 of 3 ask for mistake analysis; one student receives feedback late)	E4, E7
Chest compression is a common source of mistakes	Reported by 1 respondent (R1)	E5; also supported by literature [6]
An instructor view is needed	Reported by 1 respondent (R1)	E9; remains a planned feature (P2)
Users will accept VR	Assumption	No respondent has used VR; no one reported motion sickness (R1, R2), but without VR experience this is not evidence
Headset comfort and cybersickness are acceptable in a	Assumption	Literature reports dizziness and headaches [11]; to be measured in Section 9

15-minute session		
Compression depth can be estimated reliably with the chosen input device	Assumption	Not covered by the interviews; depends on the device [1][6]

Table 2.4. Evidence level

2.5 Limitations

- Small sample (3 respondents), convenience sampling, no information on institution or region beyond role and year.
- Written form, not live discussions: answers are short and cannot be clarified. R3 did not answer the questions about platform features and data, and some answers are general descriptions of training rather than personal experience.
- Only one instructor; students are from different specialties.
- No respondent has used VR or AR, so findings describe the problem and expectations, not the acceptance of the solution.
- Self-reported data.

2.6 Next steps

- Follow up with live discussions (short calls) with the instructor and at least one student after the prototype demo, to test the prototype scenarios (Section 8).
- Add 1-2 respondents, preferably an instructor of emergency medicine or nursing, to confirm findings E5 and E9.
- Record each new discussion in Table 2.2 and update the evidence level in Table 2.4.

Appendix A. Interview questions

- Background: current role; specialty, year of study or professional experience.
- Current training: how CPR is taught and with which tools; how blood collection and intravenous procedures are taught; number of repetitions and whether it is enough; how performance is assessed; how soon and in what way feedback is given.
- Difficulties: the hardest part; the most frequent mistakes; what prevents more practice; an example of a failed training session.
- Technology: experience with VR, AR or computer simulators and impressions; motion sickness or discomfort in VR.
- Expectations: useful features; missing features and concerns; performance data that matters; what could prevent use; anything not asked.

3. Literature review

3.1 Scope and method

- Aim of the review: to identify what is already known about AR/VR training of medical specialists, which approaches exist for feedback and assessment, and which limitations the project must take into account.
- Search sources: Google Scholar and Scopus.
- Search keywords: virtual reality, extended reality, augmented reality, mixed reality, CPR, blood draw, IV, venipuncture, medical procedures, learning analytics, training, simulation, education, software architecture.
- Period: 2021-2026; older foundational works are included only if essential.

- Selection criteria: article, review or conference paper; contains the search keywords; the abstract matches the topic. Sources that are not AR/VR or have no learning outcomes (e.g. [8], [9], [10], [12], [13]) are used as context for platform design, assessment or adoption, and are marked as indirect or partial in the table.
- Result: 13 sources used, all of them scientific publications.

3.2 Analysis table

№	Source	Year	Type of study	Procedure / domain	Approach	Technology	Participants	Main findings	Limitations	Relevance to the project
[1]	Abaszadeh et al.	2026	Systematic review (PRISMA 2020; protocol developed a priori but not registered; RoB 2, ROBINS-I, GRADE); 31 studies (19 RCTs and 12 observational, as stated by the authors), 2014-2024	Practical clinical skills in undergraduate medical education: emergency and urgent care, including BLS/CPR (12 studies); diagnostic and physical examination (10); surgical and technical skills (5); imaging and procedural guidance, including ultrasound-guided vascular access and arterial blood gas collection (4)	Quantitative studies, synthesized narratively (SWiM, no meta-analysis)	Fully immersive VR with head-mounted displays and active psychomotor practice (most often Oculus Quest 2, Oculus Rift, HTC Vive; mostly custom-developed applications)	31 studies, 4,926 undergraduate medical students as stated by the authors (sample size 12-529, median 56); five databases plus grey literature, search up to March 2024	Immersive VR improved or was non-inferior to conventional training for immediate skill performance in 19 of 23 studies (certainty low); skill retention very uncertain (6 studies, very low certainty); satisfaction consistently high (very low certainty); most consistent and robust evidence for imaging and procedural guidance (4 RCTs); strongest effects against passive comparators, mostly non-inferior to hands-on training; VR equipment impaired chest compression depth and recoil in one CPR study; repeated practice was more effective than single sessions	Methodologically limited evidence: most RCTs with some concerns or high risk of bias, small samples, substantial heterogeneity that precluded meta-analysis; sparse evidence on long-term retention, and transfer to clinical practice or patient outcomes largely absent; English-only (possible language bias); publication bias cannot be excluded, especially for subjective outcomes; potential conflicts of interest in studies involving VR developers; findings may not generalize to other virtual or extended reality technologies; reported counts are internally inconsistent (totals by domain do not match the stated number of RCTs and participants)	Directly relevant background: HMD training of procedural skills, including BLS/CPR and vascular access/blood sampling (FR-1, FR-5, FR-7, NFR-2); the CPR study where headset equipment impaired compression depth supports documenting input-device limits for depth estimation (FR-6); objective checklist scores, task time and error counts match the project's scoring and data collection (FR-8, FR-9); repeated practice supports retention (FR-12); VR is seen as a complement to hands-on training, with low certainty and no long-term data
[2]	Brunzini et al.	2021	Design methodology with pilot evaluation	<procedure: check in the paper>	Quantitative	Microsoft HoloLens, Vox Gear Plus, Unity, Vuforia, Blender	18 six-year medical students	The MR simulation showed promising potential for medical training, achieving a 94.45% success rate with HoloLens, while Vox Gear Plus performed worse due to technical limitations.	Small sample size; preliminary pilot study; Vox Gear Plus limitations prevented completion of the procedure; limited field of view on HoloLens; lack of statistical analysis of MR effectiveness.	Provides a practical example of designing and developing an MR-based medical training simulation, highlighting relevant technologies, user experience evaluation, and hardware limitations; the device-dependent results support fixing the target headset and software versions early (NFR-2, R2, R4)
[3]	Saruwatari et al.	2023	Single-center randomized (crossover) user study	Ultrasound-guided vascular access	Quantitative	HoloLens 2, HoloUs, Unity, Microsoft Mixed Reality Toolkit, ARToolKit marker-based tracking	34 (22 with and 12 without experience in ultrasound-guided vascular access); 32 analysed	Task completion time improved significantly only at the third HoloUS use on day 2 (median 11.5 s vs 18.5 s with conventional ultrasound, $p = 0.04$, all users; experts: 8.5 s vs 13.5 s, $p = 0.02$). Head adjustments were significantly lower with HoloUS in all tasks (median 1 vs 3). No significant differences in attempts, needle redirections or needle visualization rate. Novices showed the same trend, but it was not significant ($p = 0.88$). Experts rated temporal workload lower than novices (NASA-TLX, $p = 0.03$).	Single center; small subgroups; novices were mostly engineers rather than trainees; improvement appeared only after repeated use; one narrow task	Objective metrics (time, attempts, needle redirections) for scoring and data collection (FR-8, FR-9); frame rate and latency reported as quality indicators (NFR-1); the benefit appeared only after repeated use, which supports repeatable practice (FR-12)
[4]	Rochlen et al.	2022	Single-group prospective evaluation study (no control group)	Peripheral intravenous catheter (PIVC) placement	Mixed	HoloLens fixed on a stand, projecting vein anatomy onto a monitor; Unity, ARToolKit marker tracking; Laerdal mannequin arm	62 (medical students, residents, fellows, faculty, nurses, nurse anesthetists; single institution); 59 completed the two-week follow-up	First-attempt success 77.4% (48/62); extravasation 17.7%, hematoma 4.8% on first attempt; 62.9% rated MR superior to other procedural training methods (4.8% worse); 51.6% preferred MR combined with instructional videos; most participants reported improved experience in later clinical PIVC placement (self-reported)	No control group; outcomes mostly self-reported; single institution; headset was not worn (COVID-19 protocol), which changes the interaction; marker on the needle interfered with grip; some technical glitches; does not recreate handling the needle in the vein with the catheter still outside	Checklist-based real-time assessment with measurable outcomes (FR-3, FR-8, FR-9); marker-based tracking can disturb natural tool handling, a design risk for the blood draw scenario (FR-6, FR-7)
[5]	Wu et al.	2026	Multicenter parallel-group randomized controlled trial (block randomization; difference-in-differences and	Venipuncture (nursing skills)	Quantitative, with qualitative interviews as a complement	VR with 6-DoF haptic handles, multimodal sensing (hand trajectories, voice, gaze at 100 Hz), deep regression-Transformer scoring	120 vocational nursing students at three colleges (Sichuan and Chongqing); 12	Accuracy rose by 21.8 percentage points in the VR closed-loop group vs 9.4 in the control group; feedback delay fell from 38 s to 9 s ($p < .001$) and fully mediated skill transfer (standardized indirect	12 weeks only, no long-term retention data; vocational students only; depends on network bandwidth and a high-performance GPU; very complex custom system that is hard to reproduce; inconsistent cost figures	Shows that fast, objective feedback is the key mechanism for skill transfer (FR-4); system-vs-examiner agreement (Cohen's κ) is a ready method for validating automated scoring (FR-8, R5, Section 9); high hardware requirements limit deployment, which

№	Source	Year	Type of study	Procedure / domain	Approach	Technology	Participants	Main findings	Limitations	Relevance to the project
			structural equation modeling for analysis)			network; GPU workstation (NVIDIA RTX A4000) supporting 8 VR terminals	weeks; OSCE at weeks 1, 6 and 12	effect = -0.31); agreement between system and examiner scores was κ = 0.81 (vs 0.64 in the traditional group); costs about 27-30% lower	within the paper (hardware cost vs. cloud GPU cost, 27% vs. 30% savings); part of the scoring relies on the system's own evaluation	supports keeping the prototype lightweight on a standalone headset (NFR-1, NFR-2, R7)
[6]	Sabery et al.	2026	Scoping review; 31 experimental, quasi-experimental or RCT studies, 2006-2022 (the reference title says scoping review)	Cardiopulmonary resuscitation (CPR), chest compression quality	Quantitative studies, synthesized narratively (no meta-analysis)	Real-time feedback systems: visual, auditory, audiovisual, mobile apps, smartwatches, gamification, AR/VR, video-based (e.g. TrueCPR, Mini-VREM)	Medical students, healthcare professionals and laypersons across the included studies (31 studies; databases: PubMed, Web of Science, Scopus)	Real-time feedback generally improved compression depth, rate, full recoil and hand position (e.g. adequate rate from ~70% to >82%, adequate depth from 35.9% to 54.8% in one study); effects depend on device type and learner group: strong for novices, limited or negative for experienced providers; some accelerometer-based or AED-integrated devices reduced compression depth; AR/VR results were mixed; combining video with verbal feedback gave the best retention; human assessment agreed poorly with software-based evaluation	Narrative synthesis only; included studies are simulation-based, so clinical outcomes are unknown; English-only; search period ended in 2024 (the paper gives two different search dates, 19 Dec and 16 Sep 2024); feedback sometimes increased workload and distraction; retention and transfer to real practice are not established	Supports real-time feedback as a core feature (FR-4) and measurement of rate, count and depth for CPR (FR-5, FR-6); poor human-vs-software agreement shows why scoring must be validated against clear reference values (FR-8, R5, Section 9); depth estimation can be unreliable depending on sensors (FR-6); mixed AR/VR results mean the benefit of the prototype must be evaluated, not assumed (Section 9); booster sessions motivate repeated practice (FR-12)
[7]	Mese et al.	2026	Systematic review	Hygiene and infection prevention education (hand hygiene, PPE use, surgical site infection prevention, oral hygiene)	Mixed	XR: immersive VR with headsets (53% of studies), AR via Kinect or smartphones (29%), MR	Healthcare workers (26%), K-12 students (21%), nurses, medical students, care home staff, food handlers, special groups across the included studies	Most studies report improved practical skills, behaviors or attitudes, and some report better retention and real-time feedback than traditional training; a smaller number found limited or no gains, especially in knowledge and compliance; in one included study a summary visualization of errors improved performance while real-time visualization had a negative effect	Cost, safety, validation, and developing robust frameworks for curriculum integration and policy support	Confirms the typical barriers for XR training: cost, hardware variability, cybersickness, shared-headset hygiene and lack of long-term validation (R2, R7); recommends XR as a complement to, not replacement of, traditional training; the finding on real-time vs. summary feedback suggests testing how feedback is presented (FR-4, FR-10)
[8]	Kim & Kim	2026	Quasi-experimental, non-equivalent control group, time-lagged pre-post design (non-randomized)	Disaster nursing, recovery phase (temporary shelter setup and operation, needs assessment, psychological first aid); not a clinical procedure	Quantitative	Browser-based multi-user metaverse platform on PC/laptop, no HMD or controllers; role-based avatars; custom (patented) system; 110-min session with PEARLS debriefing	76 analysed (43 experimental, 33 control) out of 168 recruited; 3rd and 4th-year nursing students from three colleges in Korea	Compared with the handout-only control group, the metaverse group improved more in disaster readiness ($p = .002$, $r = 0.72$), learning self-efficacy ($p < .001$, $r = 0.48$) and perceived importance of disaster nursing education ($p = .044$); satisfaction was high (4.57/5, SD 0.59); disaster-related attitudes did not change significantly in either group; recovery-related readiness items showed large effects in the experimental group (Cohen's $d = 0.978, 1.014, 0.976$), while the control group showed no gain in information collection and analysis	Self-report outcomes (response bias, may not reflect objective performance); non-randomized assignment by class schedule (selection bias); time-lagged non-randomized design (maturation and expectancy effects); sample smaller than planned with unequal groups (reduced power, Type II error, overestimated effect sizes); attrition due to non-consent (may affect group comparability); limited number of nursing schools in specific regions of Korea (limited generalizability); results are preliminary evidence, not definitive proof of superiority	Supporting rather than key source: indirect relevance, since the topic is disaster-recovery readiness of nursing students, not a clinical procedure, and the platform is browser-based (PC, no headset), so it is not AR/VR in the strict sense; it supports the general claim that simulation improves readiness and self-efficacy of health-profession trainees and that low hardware requirements can still be effective (NFR-2, R7); outcomes are self-reported only, with no objective performance metrics, which justifies automated scoring of step accuracy and timing in the project (FR-8, FR-9); structured pre-briefing and debriefing is a model for the end-of-session review (FR-10); the weak handout-only control group shows why the project uses guideline reference values and expert assessment as its baseline (Section 9)
[9]	Benammi et al.	2026	Quasi-experimental pretest/posttest study with a control group (post-test only in the control group)	Basic care procedures for first-year medical students: hand hygiene and antisepsis, vital sign monitoring, injection techniques, vascular access, blood sampling	Quantitative	Web-based e-learning platform MedTuto (WordPress, Tutor LMS); simulation-based instructional video capsules (mannequins, close-up of hands and instruments), technical sheets, PowerPoint	240 first-year medical students, Faculty of Medicine, Pharmacy, and Dentistry of Fez (Morocco): 120 experimental, 120 control; 3-week immersion internship in basic care	Experimental group improved from 13.51 to 16.14 points (+2.63, $p < 0.001$, Hedges' $g = 0.64$); posttest was higher than control (13.54) by 2.60 points ($p < 0.001$, $g = 0.82$); score dispersion decreased, and students with lower baseline scores improved more; questionnaire reliability Cronbach's $\alpha = 0.758$	Evaluation focused on cognitive outcomes (MCQs), not psychomotor performance; results are not direct evidence of clinical competence; short-term assessment, no conclusions on long-term retention; single institution, limited generalizability	Indirect: video-based platform, not AR/VR, but covers the same procedures (vascular access, blood sampling); separate module per procedure with a defined scenario supports the procedure module format (FR-11, NFR-8); step-by-step technical sheets with safety principles and expert review of the test instrument support guideline-traceable steps and expert validation (FR-7,

№	Source	Year	Type of study	Procedure / domain	Approach	Technology	Participants	Main findings	Limitations	Relevance to the project
						materials, MCQ quizzes; no AR/VR				NFR-5, R1); knowledge-only MCQ assessment without psychomotor metrics justifies step-level checking and scoring (FR-3, FR-8, FR-9)
[10]	Cazorla-Calderón et al.	2026	Qualitative descriptive study with a contextual quantitative component (three focus groups, template analysis; ad hoc questionnaire)	Integration of emerging technologies in undergraduate nursing education, from the faculty perspective; not a specific clinical procedure	Qualitative	No specific system: high-fidelity simulation, VR, AR, MR, XR and AI tools (e.g. generative AI chatbots) discussed in general	22 nursing faculty from three universities in Spain; three focus groups (n = 9, 7, 6); 7-item questionnaire on technology experience (Cronbach's $\alpha = 0.85$)	Three themes: Resources and Integration, Teaching and Learning, Navigating the Digital Divide; faculty see ET as pedagogically valuable, but report limited institutional support, cost, large cohorts (350-500 students), unequal training and fear of losing humanistic and relational aspects of care; high-fidelity simulation was the most familiar technology, MR and AR the least; low-cost technologies were seen as a complement to, not a replacement for, theoretical instruction; adoption depended more on institutional culture, resources and prior exposure than on age	Relatively small sample from a limited number of institutions (limited transferability); faculty perspective only, no students; interpretation shaped by researchers' perspectives (mitigated by reflexivity and triangulation); template analysis can underrepresent atypical viewpoints and gives no causal claims; quantitative component descriptive only	Indirect: faculty perceptions, not a training system; shows the main adoption barriers (cost, hardware and technical support, large cohorts, training, risk of losing the human element), which supports a lightweight prototype on one standalone headset (NFR-2); XR is seen as a complement to traditional teaching
[11]	Raurell-Torredà et al.	2026	Single-center, open-label randomized controlled trial (1:1, groups of 6-8 students randomized), with thematic analysis of open-ended comments	Basic CPR (AED, chest compressions, ventilations, ABCDE assessment) in undergraduate nursing training	Mixed (mainly quantitative)	Immersive VR: Meta Quest 3 headsets and controllers, Warp platform (Immersium Studio); 15-min individual scenario in a primary care room, decision-making along the CPR algorithm, feedback from the system; followed by high-fidelity manikin simulation (Laerdal)	246 4th-year nursing students randomized (single Spanish public university, 2024-25); 211 analysed (107 experimental: VR + manikin simulation; 104 control: manikin simulation only; control group received VR after the evaluation)	No significant differences in self-confidence (adjusted effect 0.02, $p = 0.974$) or group CPR performance (adjusted effect 0.56, $p = 0.153$); experimental group rated usability higher (usefulness, ease of use, ease of learning, satisfaction, all $p \leq 0.002$), but questionnaire timing differed between groups; side effects: dizziness 33.6% "sometimes" and 13.7% "most of the time", headache 19% and 9.5%, nausea 7.6% and 2.4%; ambiguity tolerance predicted higher self-confidence (OR 1.083, $p = 0.002$); students criticized missing right/wrong feedback, no way to go back or ask questions, heavy headset and cost	Underpowered for CPR performance because it was assessed at group level (21 vs 18 groups); no individualized CPR assessment of students; inter-rater reliability not assessed due to limited human resources (evaluators trained in a pilot test); single center, results extrapolable only to similar simulation-based teaching contexts; USE questionnaire timing and VR exposure differed between groups (possible order or novelty effect on usability ratings)	Directly relevant: immersive VR for CPR on a standalone headset (FR-1, FR-3, NFR-2); students' complaints about missing correct/incorrect feedback and no way to go back support step checking, immediate feedback and repeating a procedure (FR-3, FR-4, FR-12); CPR scored only at group level shows why per-step, individual metrics are needed (FR-8, FR-9); the scenario script was written by anesthesiologists and critical care nurses, a model for expert validation (NFR-5, R1); dizziness and headaches support comfort requirements (NFR-4), and the 15-min session limit supports NFR-9
[12]	Lu et al.	2026	Design study / system description (SIGGRAPH Immersive Pavilion short paper); no user evaluation	Laboratory mouse blood collection (gripping, body positioning, selection of blood sampling site); veterinary and biomedical training, not human clinical care	Qualitative / descriptive	Immersive VR: head-mounted display and handheld controllers, animated 3D mouse model, laboratory environment; real-time assessment of hand placement, grip location and site selection; visual, auditory and vibrotactile feedback; game-based staged progression with repeatable attempts and a visual hint for the puncture site	None reported (no trainees or participants described)	No measured outcomes. The authors describe a system that checks each step against predefined criteria from standard laboratory practice, gives immediate feedback and corrective guidance, and requires the user to correct an error before proceeding; they argue that it supports early skill acquisition, repeatable risk-free practice and the Replacement and Refinement principles of the 3Rs	Not stated by the authors as a separate section; they list as future work expansion to more procedures, performance metrics for competency assessment and evaluation of training outcomes in blended curricula, which implies that no performance metrics or outcome evaluation exist yet	Directly relevant by design: an immersive VR trainer with a procedure-specific step check, immediate multimodal feedback, forced correction before the next step, repeatable practice and staged progression (FR-2, FR-3, FR-4, FR-12); its future work (performance metrics for competency assessment) matches the project's scoring and data collection (FR-8, FR-9); no evidence of effectiveness, so it is a design reference, not support for learning outcomes
[13]	Jain et al.	2025	Design and development study with technical calibration experiments and a small pilot validation	Endotracheal intubation / difficult airway management with a video laryngoscope (emergency and critical care)	Mixed	Hybrid physical-virtual VR system: HTC Vive Pro, Unity 3D, 3D-printed anatomical parts, AirSim Advance X manikin; laryngoscope and stylet	Positional calibration: 10 subjects (6 trials each); parameter calibration: 7 expert physicians (>100 intubations, >5 years	Positional calibration accuracy within 0.1 cm; pressure profiles of the VR system did not differ significantly from the C-MAC video laryngoscope (Hotelling T^2); clinicians rated feasibility,	No separate limitations section; the authors state that integrated tools for automated performance assessment are missing, that the study mainly relies on users' self-assessment of realism and accuracy, and that further	Partial: intubation, not CPR or blood draw, and a tethered setup with custom sensors, unlike a standalone headset (NFR-2); separate modules (laryngoscopy, vital signs, tube insertion) and scenarios defined in JSON/CSV are a reference for the procedure

№	Source	Year	Type of study	Procedure / domain	Approach	Technology	Participants	Main findings	Limitations	Relevance to the project
			(TAM survey); no control group, no learning outcomes			with pressure sensors (Arduino Nano BLE); Hexoskin biometric vest; Pulse physiology engine for vital signs (scenarios set via JSON/CSV); real-time soft-tissue deformation and haptic feedback; sensor data streamed at 10 Hz	of experience); TAM survey: 7 clinicians; one medical training center (University of Arizona)	perceived usefulness and ease of use positively; feedback: virtual hands should be visualized and soft-tissue deformation sensitivity adjusted	work is needed to comprehensively assess effectiveness (objective metrics, comparative studies)	module and definition format (FR-11, NFR-8); requirements were set by clinical experts together with end users, a model for expert validation (NFR-5, R1); per-session performance data collection (FR-9); the authors name missing automated performance assessment as a gap, which the project's scoring addresses (FR-8)

Table 3.1. Literature review matrix(academic sources only)

3.3 Critical review

3.3.1 Effectiveness of AR/VR in medical skills training

The strongest overview is the systematic review [1] (31 studies, 4,926 participants as stated by the authors): immersive VR improved or was non-inferior to conventional training for immediate skill performance in 19 of 23 studies, but the certainty of evidence was low, and very low for retention and satisfaction. Effects were largest against passive comparators and mostly non-inferior against hands-on training. Primary studies agree only partly. The multicenter RCT [5] reported a larger accuracy gain with VR (+21.8 vs +9.4 percentage points), whereas the RCT [11] (246 students) found no difference in self-confidence or group CPR performance, and in [3] the time benefit of AR appeared only after repeated use and most other metrics did not differ. Studies [4], [8] and [9] have no control group, non-randomized groups or knowledge-only and self-reported outcomes, so they show acceptance and knowledge gains rather than skill gains. Overall the evidence is promising but weak: small and single-center samples, short follow-up, mostly nursing students, and few objective measures. VR is therefore treated as a complement to hands-on training, not a replacement.

3.3.2 Feedback and objective performance assessment

Real-time feedback is the most consistent mechanism in the sources. The review [6] found that it improves CPR depth, rate, recoil and hand position, mostly for novices, and that human assessment agreed poorly with software-based evaluation. In [5] a shorter feedback delay (38 s to 9 s) mediated skill transfer, and system scores agreed well with examiners ($\kappa = 0.81$). The review [7] reports that a summary of errors helped while real-time visualization hurt in one study, so the way feedback is presented matters. Several systems check steps against predefined criteria and require correction before the next step [12], or use checklists and sensors for time, attempts and tool pressure [3], [4], [13]. Assessment is often weak, however: [11] scored CPR only at group level and without inter-rater reliability, students there complained that the VR scenario did not show whether an answer was right, [8] and [9] used self-report and MCQs only, and [13] states that automated performance assessment is still missing.

3.3.3 Training of specific procedures (CPR, blood draw / venous access)

For CPR, the evidence comes from the RCT [11], the feedback review [6] and the CPR studies inside [1]. These show that VR can support the CPR algorithm and decision-making, but that headset equipment can impair compression depth and recoil [1], and that depth estimation depends on the input method and sensors [6]. For blood draw and venous access, the sources use AR/MR for ultrasound-guided vascular access [3] and peripheral catheter placement [4], VR with haptic handles and automated scoring for venipuncture [5], a step-checking VR trainer for mouse blood collection [12], and video modules for vascular access and blood sampling [9]. The domain on ultrasound-guided access and arterial blood gas collection in [1] gave the most consistent results.
Input methods differ widely (controllers, haptic handles, marker tracking), and

marker tracking interfered with natural tool handling in [4]. No source covers CPR and blood draw in one standalone-headset application.

3.3.4 Platform design and extensibility

Most sources describe a single custom system for one procedure: [12] for mouse blood collection,
[13] for intubation, [5] for venipuncture and [11] for CPR. Their design lessons are still useful. [13] splits the system into modules (laryngoscopy, vital signs, tube insertion) and defines physiological scenarios in JSON/CSV files; [9] organizes content as one module per procedure with a defined scenario (objectives, steps, context, roles); [2] shows that the choice of device can decide whether a procedure can be completed at all. The system in [5] needs a GPU workstation and is hard to reproduce, which argues for a lightweight design. None of the sources describes a common, documented procedure format that allows new procedures to be added without changing the core system.

3.3.5 Adoption, usability and educators' perspective

Reported barriers are cost, hardware and technical support, large cohorts, training of educators, and the risk of losing human and relational aspects of care [10], [7]. Learners rated VR as useful and easy to use [11], but 33.6% reported dizziness "sometimes", 19% headaches, and they mentioned heavy headsets, technical problems and cost [11]; simulator sickness is also a listed feasibility issue in [1]. Browser-based and low-hardware solutions lower cost but still depend on bandwidth and device access [8], and high-end setups limit deployment [5]. Educators and reviews agree that XR should complement traditional teaching [10], [7], [1], and that instructor involvement and repeated practice are needed.

3.4 Research gap and implications for the project

- Gap: the reviewed solutions are mostly single-procedure systems tied to a specific device or setup, no common extensible procedure format is described, objective step-level scoring is rarely implemented or validated, and the evidence is mostly short-term, small-scale and based on nursing students, while the project targets medical students and interns.
- Implication 1: the project defines procedures in a procedure module with steps, conditions, metrics and scoring rules, and adds the second procedure without changing the core system (FR-11, NFR-8).
- Implication 2: the project implements step checking, immediate feedback and per-step data collection, and validates the scoring against guideline values and expert assessment (FR-3, FR-4, FR-8, FR-9, R5).
- Implication 3: the project takes into account input-device limits for CPR depth, comfort and session length, and the lightweight standalone-headset setting (FR-6, NFR-1, NFR-2, NFR-4, NFR-9, R2, R7), and traces every step to a published guideline with expert review (NFR-5, R1, R10).

3.5 Summary

Existing work shows that immersive and mixed reality can support procedural training, that fast and objective feedback is the most consistent success factor, and that VR works best as a complement to hands-on practice. The evidence is of low certainty, with small samples, short follow-up, and few objective or validated measures; side effects and cost are recurring barriers. Most solutions cover a single procedure on a specific device, and none describes a common extensible procedure format.
This review therefore shapes the requirements for step guidance, immediate feedback, per-step scoring and data collection (FR-2 to FR-10), an extensible procedure module (FR-11, NFR-8), and a lightweight, comfortable prototype (NFR-1, NFR-2, NFR-4). It also shapes the verification plan, which compares sessions against guideline reference values and expert assessment (Section 9).

4. Existing System Comparison

This section compares four existing commercial VR training systems against eight criteria. Selection rationale: the systems are used in medical education, run on VR headsets, and cover different training domains (surgery, clinical scenarios, resuscitation), so together they show what the market offers. Information is taken from vendor materials and independent reviews listed in References; where a source does not state a capability, the cell says "not stated" instead of assuming.

4.1 Systems reviewed

- Osso VR: VR surgical and procedural training platform with integrated analytics [24][26].
- Oxford Medical Simulation (OMS): VR clinical scenario simulation for nursing and medical education [28][29].
- Health Scholars: voice-driven VR training for Advanced Cardiac Life Support (ACLS) [30][31].

- Fundamental Surgery (FundamentalVR): VR surgical simulation platform with haptic feedback [32][33].

4.2 Comparison

Criterion	Osso VR	Oxford Medical Simulation	Health Scholars (ACLS)	Fundamental Surgery	Proposed project
1. Technology	VR headset [26]	Screen-based or VR headset [28]	VR headset [30]	VR, with haptic devices in the HapticVR mode [32][33]	VR (Unity, OpenXR)
2. Target hardware	Standalone headsets such as Meta Quest [24]; dedicated headsets and physical space required [27]	Headset-based; Quest 2 used in an NHS pilot [29]	Meta Quest 2 [30]	Standalone headsets (Quest, Vive Focus Plus) or PC with headset and haptic arms [32]	Meta Quest 3 (Quest 2 compatible)
3. Procedures covered	Surgical and medical procedures, orthopedics library [26]	200+ clinical scenarios, including paediatrics, emergencies and bedside procedures [29]	ACLS scenarios; learner directs a virtual team to shock, give medication or perform CPR [30]	Orthopedic surgical procedures [34]	CPR and blood draw (prototype); more procedures planned
4. Hands-on basic skills (CPR compressions, blood draw)	Not stated	Not stated	Not stated; CPR is directed by voice, not performed by the learner [30]	Not stated	Yes, both are implemented in the prototype
5. Customization and extensibility	Custom modules developed by the vendor [25]; a reviewer notes limited customization for institution-specific protocols [27]	Authoring platform (OMS Create) for modifying or building scenarios [28]	Not stated	Not stated	Procedures added as independent modules by the institution
6. Automated assessment and feedback	Integrated assessment and analytics [26]	Personal feedback, guided reflection and scores [28]	Validates competencies via the simulation [30]	Learner progress monitored at each surgical step [33]	Basic feedback in the prototype (to be aligned with Section 6)
7. Instructor view and analytics	Analytics and reports at individual and organizational level [26]	Analytics for administrators, faculty and learners [28]	Not stated	Remote instructor dashboard [33]	Deferred to pre-defense 2
8. Licensing model	Commercial, enterprise platform [24]	Commercial: pay per use, library or tiered pricing, contact vendor [28]	Commercial, pricing not stated	Commercial SaaS [34]	Developed in-house, institution-owned

Table 4.1.Existing system matrix

4.3 Analysis

- All four reviewed systems are proprietary commercial products. Each source

describes either vendor-built content [25], a vendor authoring tool [28], or does not mention customization.

- Coverage is concentrated in surgery [26][34], clinical decision-making scenarios [28] and ACLS team leadership [30]. The reviewed sources do not state support for hands-on basic skills such as CPR compressions or blood draw.
- Academic prototypes in the literature review address individual procedures: CPR training with immersive VR [11], blood collection [12] and airway management [13]. They do not offer a multi-procedure platform.
- Analytics and instructor views are common among commercial systems [26][28][33]. In the proposed project this capability is planned for pre-defense 2, so it is currently a gap relative to the competitors.

4.4 Conclusion

4.4 Conclusion

Within the reviewed sources, no system combines (a) hands-on basic procedures (CPR, blood draw),
(b) institution-controlled extensibility and (c) a low-cost standalone headset target. This is the niche for the proposed platform. Limitations of this comparison: only four systems were reviewed, much of the evidence comes from vendor materials, and "not stated" means the capability was not found in the sources, not that it is absent.

5. Technology Comparison

This section compares alternatives for four key technical decisions. Each decision is evaluated against criteria derived from the requirements specification (Section 6). Decisions 1-3 are applied in the pre-defense 1 prototype. Decision 4 is a preliminary choice, planned for pre-defense 2.

5.1 Summary of decisions

#	Decision	Alternatives considered	Selected	Status
1	Application engine	Unity [14]; Unreal Engine [17]; WebXR [18]	Unity	Applied in prototype
2	XR interaction layer	XR Interaction Toolkit + OpenXR [14][15]; Meta XR SDK [16]; SteamVR (OpenVR) [19]	XR Interaction Toolkit + OpenXR	Applied in prototype
3	Target device	Meta Quest 2/3 [16]; PCVR (SteamVR headsets) [19]; Pico 4 [20]	Meta Quest 3 (Quest 2 as compatible fallback)	Selected, hardware not yet available
4	Backend and result storage	Local storage (JSON/SQLite); Spring Boot + PostgreSQL [21][22]; Firebase [23]	Spring Boot + PostgreSQL (REST)	Preliminary, pre-defense 2

Table 5.1. Technology summary

5.2 Decision 1: Application engine

Requirement drivers: multi-procedure extensibility, 3D interaction and physics, development time, team experience, performance on standalone headsets.

Criterion	Unity [14]	Unreal Engine [17]	WebXR (three.js / A-Frame) [18]
Standalone headset support	Supported for Quest via OpenXR [14][16]	Supported for XR devices [17]	Via browser, performance depends on the browser and device [18]
Extensibility (new procedures as scenes/modules)	High (scenes, prefabs, ScriptableObjects) [14]	High (levels, Blueprints) [17]	Medium (manual structure)
Learning curve and team experience	Low: C#, existing team experience	High: C++/Blueprints, no team experience	Medium: JavaScript, no built-in physics engine
Physics and interaction tooling	Built in, plus XR Interaction Toolkit [14]	Built in [17]	Requires third-party libraries [18]
Build size and hardware requirements	Moderate	Higher	Lowest, but browser-dependent
Asset ecosystem for 3D content	Large (Asset Store) [14]	Large (Fab marketplace) [17]	Small

Table 5.2. Application engine comparison

Decision: **Unity**. It fits the extensibility requirement (each procedure is an independent scene/module), matches existing team experience, and the prototype is already built on it. Unreal offers higher visual fidelity but adds learning cost and heavier builds without a corresponding requirement. WebXR is easy to distribute but has limited performance and tooling for complex interactions [18].

5.3 Decision 2: XR interaction layer

Requirement drivers: hand/controller interaction, hardware independence, ability to develop without a headset.

Criterion	XR Interaction Toolkit + OpenXR [14][15]	Meta XR SDK [16]	SteamVR (OpenVR) [19]
Vendor independence	High (OpenXR is an open standard) [15]	Low (Meta devices) [16]	Medium (PC headsets) [19]
Standalone Quest support	Yes, via OpenXR plugin [14][16]	Yes, native [16]	No (PC only) [19]
Development without headset	Yes (XR Device Simulator) [14]	Limited	Limited
Ready-made interactions (grab, poke, UI)	Yes, built in [14]	Yes [16]	Yes [19]
Future portability to other headsets	High [15]	Low [16]	Medium [19]

Table 5.3. XR interaction layer comparison

Decision: **XR Interaction Toolkit + OpenXR**. Hardware independence supports the platform's long-term extensibility [15], and the XR Device Simulator allows development and testing without access to a physical headset [14], which is a current project constraint.

5.4 Decision 3: Target device

Requirement drivers: no dependency on a powerful PC, accessibility for a university, OpenXR support, cost.

Criterion	Meta Quest 2/3 [16]	PCVR (SteamVR headsets) [19]	Pico 4 [20]
Standalone operation	Yes [16]	No, requires a capable PC [19]	Yes [20]
Hardware cost	Moderate	High (headset plus PC)	Moderate
OpenXR support	Yes [16]	Yes [15][19]	Yes [20]
Developer ecosystem and documentation	Large [16]	Large [19]	Smaller [20]
Suitability for classroom deployment	High (no PC per station)	Low (PC per station)	Medium

Table 5.4 Target device

Decision: **Meta Quest 3** (Quest 2 compatible). Standalone operation removes the need for a PC per training station [16], and the developer documentation covers Unity and OpenXR workflows [16].

Because the choice goes through OpenXR [15], the application is not locked to a single device. Limitation: performance measurements on the real device are deferred until hardware access is available.

5.5 Decision 4: Backend and result storage (preliminary)

Planned for pre-defense 2. Required once roles, persistent result storage and the instructor view are introduced.

Requirement drivers: instructors view results of multiple trainees, role-based access, structured result data, team experience.

Criterion	Local storage (JSON/SQLite)	Spring Boot + PostgreSQL [21][22]	Firebase [23]

Shared results across devices (instructor view)	No	Yes [21][22]	Yes [23]
Role-based access control	Not supported	Spring Security [21]	Firebase Authentication and security rules [23]
Structured/relational result queries	Limited	Strong (SQL) [22]	Document-based (NoSQL) data model [23]
Team experience	High	High (Java Spring Boot)	Low
Control over data and hosting	Full, but local only	Full	Vendor-hosted, vendor lock-in [23]
Implementation effort	Lowest	Moderate	Low to moderate

Table 5.5. Backend + database (preliminary)

Decision: **Spring Boot + PostgreSQL (REST API)**. It supports the instructor view and roles [21], fits relational data (users, procedures, sessions, step-level results) [22], and matches existing team experience. Local storage is retained as a prototype-stage fallback until the backend is introduced.

5.6 Conclusion

The selected stack (Unity, XR Interaction Toolkit with OpenXR, Meta Quest as the target device, and a planned Spring Boot + PostgreSQL backend) is justified by the extensibility, hardware-independence and team-experience requirements. Choices for the engine, interaction layer and device are already reflected in the prototype. The backend choice is preliminary and will be confirmed at pre-defense 2.

6. Requirements specification

6.1 Approach

- Requirements are derived from the project proposal (Section 1), the literature review (Section 3), the comparison of existing systems (Section 4) and the identified risks (Section 10.5).
- **Priority** (MoSCoW): Must = required for the prototype; Should = important, can be dropped under time pressure; Could = optional.
- **Stage:** P1 = prototype for pre-defense 1; P2 = planned for pre-defense 2.
- **Source** shows where the requirement comes from. "Assumption" means it is not yet confirmed by interviews (Section 2) and is treated as unverified.
- Every requirement has an acceptance criterion that can be checked by a test, a demonstration or an inspection.

6.2 Functional requirements

ID	Requirement	Priority	Stage	Acceptance criterion	Source
FR-1	The user can choose a procedure (CPR or blood draw) in a menu at application start	Must	P1	The menu lists both procedures; selecting one starts the corresponding scenario without restarting the application	Proposal 1.6

FR-2	The system guides the user step by step through the procedure	Must	P1	For every step in the procedure definition, an instruction is shown when the step becomes active (checked by walking through all steps of both procedures)	Proposal 1.4 (obj. 6)
FR-3	The system checks whether each step was performed correctly and in the correct order	Must	P1	In a scripted correct run all steps are accepted; in at least one scripted incorrect run (wrong action or wrong order) the step is not accepted	Proposal 1.4 (obj. 6, 7)
FR-4	The system gives immediate feedback on correct and incorrect actions	Must	P1	Visual or audio feedback appears within 1 second after the action (measured in a test run)	Proposal 1.4 (obj. 6); Lit. review [6]
FR-5	CPR scenario: the system measures compression rate and counts compressions and compares them with guideline values	Must	P1	(a) Measurement accuracy: in a test session of at least 30 compressions, the measured count equals the reference count and the measured rate is within ±5 compressions/min of the reference timing. (b) Guideline check: the system shows whether the user's rate is inside the AHA/ERC range (100–120/min)	Proposal 1.4 (obj. 7); Lit. review [6]
ID	**Requirement**	**Priority**	**Stage**	**Acceptance criterion**	**Source**
FR-6	CPR scenario: the system estimates compression depth from the input method	Should	P1	Depth is displayed and compared with the guideline range, or the limitation of the input device is documented; if displayed, depth is also recorded (see FR-9)	Assumption; Lit. review [1], [6]
FR-7	Blood draw scenario: the system tracks the sequence of procedure steps defined from a published guideline	Must	P1	All steps of the blood draw checklist are implemented and traceable to the guideline (see NFR-5)	Proposal 1.6
FR-8	The system uses a scoring rule for each step defined in the procedure definition	Must	P1	Each step has a rule (at least pass/fail); the session score is computed from these rules and matches a manual calculation for a recorded session	Proposal 1.6; Risk R5
FR-9	The system collects performance data during a session (time per step, step correctness; for CPR also compression rate and count)	Must	P1	After a session, the recorded data contains time and result for every step; for the CPR scenario it also contains rate and count	Proposal 1.4 (obj. 7); Lit. review [3], [13]
FR-10	At the end of a session the system shows a results summary (total time, completed, missed and incorrect steps, score)	Must	P1	The summary is shown after both procedures and matches the recorded data	Proposal 1.5
FR-11	New procedures are added through a procedure module (steps, conditions, metrics, scoring rules) without changes to the core system	Must	P1	The second procedure (blood draw) is added through the module; the Git diff shows no changes to core code	Proposal 1.4 (obj. 4, 8); Lit. review [9], [13]
FR-12	The user can repeat a procedure and return to the	Should	P1	After the summary, the user can restart the same procedure or return to the menu	Proposal 1.1 (repeated

	menu after a session				practice)
FR-13	User roles: trainee, instructor, administrator	Should	P2	Different roles see different functions after login	Proposal 1.2, 1.6
FR-14	Persistent storage of session results and progress tracking	Should	P2	Results of a session are available after the application is restarted	Proposal 1.6
FR-15	Instructor view for reviewing trainee results	Should	P2	An instructor can open the list of sessions of a trainee and see their summaries	Proposal 1.2, 1.6

Table 6.1. Functional requirements

6.3 Non-functional requirements

ID	Category	Requirement	Priority	Stage	Acceptance criterion	Source
NFR-1	Performance	The application runs at a stable frame rate on the target headset	Must	P1	Average frame rate is at least 90% of the headset's native refresh rate during a 5-minute session (profiler), with no drop below that level longer than 1 second	Risk R2; Lit. review [3]
NFR-2	Compatibility	The prototype runs as an application on the selected headset with fixed Unity and XR package versions	Must	P1	The built application starts and both procedures can be completed on the target device	Risks R4, R7; Lit. review [2]
NFR-3	Usability	A new user can start and complete a procedure after a short introduction (smoke usability check, not a statistical evaluation)	Should	P1	3 or more test users complete a procedure after at most 2 minutes of instruction; the SUS score is reported as indicative only	Assumption; Lit. review [11]
NFR-4	Comfort	The application does not move the user's view without the user's action (no forced locomotion)	Should	P1	Inspection of both scenarios confirms no artificial camera movement	Assumption; Lit. review [11]
NFR-5	Medical accuracy	Every procedure step is traceable to a published clinical guideline	Must	P1	Each step in the procedure definition has a guideline reference; unvalidated steps are marked as such	Risks R1, R3
NFR-6	Privacy	The prototype does not collect personal data; results are stored only locally or anonymised	Must	P1	Inspection of stored data shows no personal identifiers	Risk R8
NFR-7	Reliability	A full session completes without crashes	Should	P1	10 consecutive full sessions per procedure without crash	Assumption
NFR-8	Extensibility	The procedure definition format is documented so that a new procedure can be described without reading core code	Should	P1	The format is described in the documentation (Section 7) and was used for the second procedure	Proposal 1.4 (obj. 4)

NF R-9	Comfort	A single session does not exceed 15 minutes	Could	P1	Measured duration of a full run of each procedure is at most 15 minutes	Assumption; Lit. review [11]

Table 6.2. Non-functional requirements

6.4 Out of scope

Requirements for additional procedures, clinical validation with real patients, support for all AR/VR devices and integration with learning management systems are out of scope (see Section 1.6).

7. Architecture and diagrams

7.1 Overview

The prototype (P1) is a single Unity application for a standalone XR headset. It contains a core that is independent of any procedure and a procedure module that describes each procedure (steps, conditions, metrics, scoring rules). CPR and blood draw are two procedure modules loaded by the same core, so the second procedure can be added without changes to the core (FR-11, NFR-8).
Roles, persistent storage and the instructor view (P2) are shown as planned components and are not part of the P1 prototype.

Main design decisions:

- **Procedure as data.** Steps, conditions, scoring rules and guideline references are stored in a procedure definition file, not in core code (FR-11, NFR-5, NFR-8).
- **Input is isolated.** The Input Adapter converts controller or hand-tracking input into device-independent action events, so the device decision (Section 5) does not change the core (FR-6, NFR-2).
- **Metrics are pluggable.** The core defines a metric interface. Continuous CPR metrics (rate, count, depth) are implemented as metric providers that belong to the CPR module, not to the core (FR-5, FR-6, FR-9).
- **Local anonymised storage in P1.** Session results contain no personal identifiers (NFR-6). Persistent storage and accounts are P2 (FR-14).

7.2 System context diagram

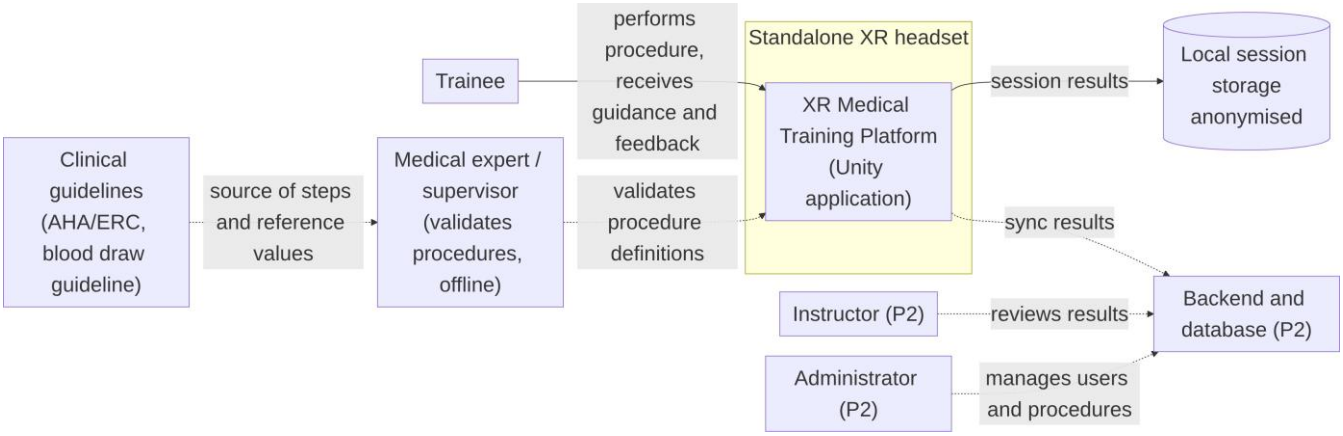


Fig 7.1. System context diagram

Solid lines are P1, dotted lines are offline processes or P2.

7.3 Use case diagram

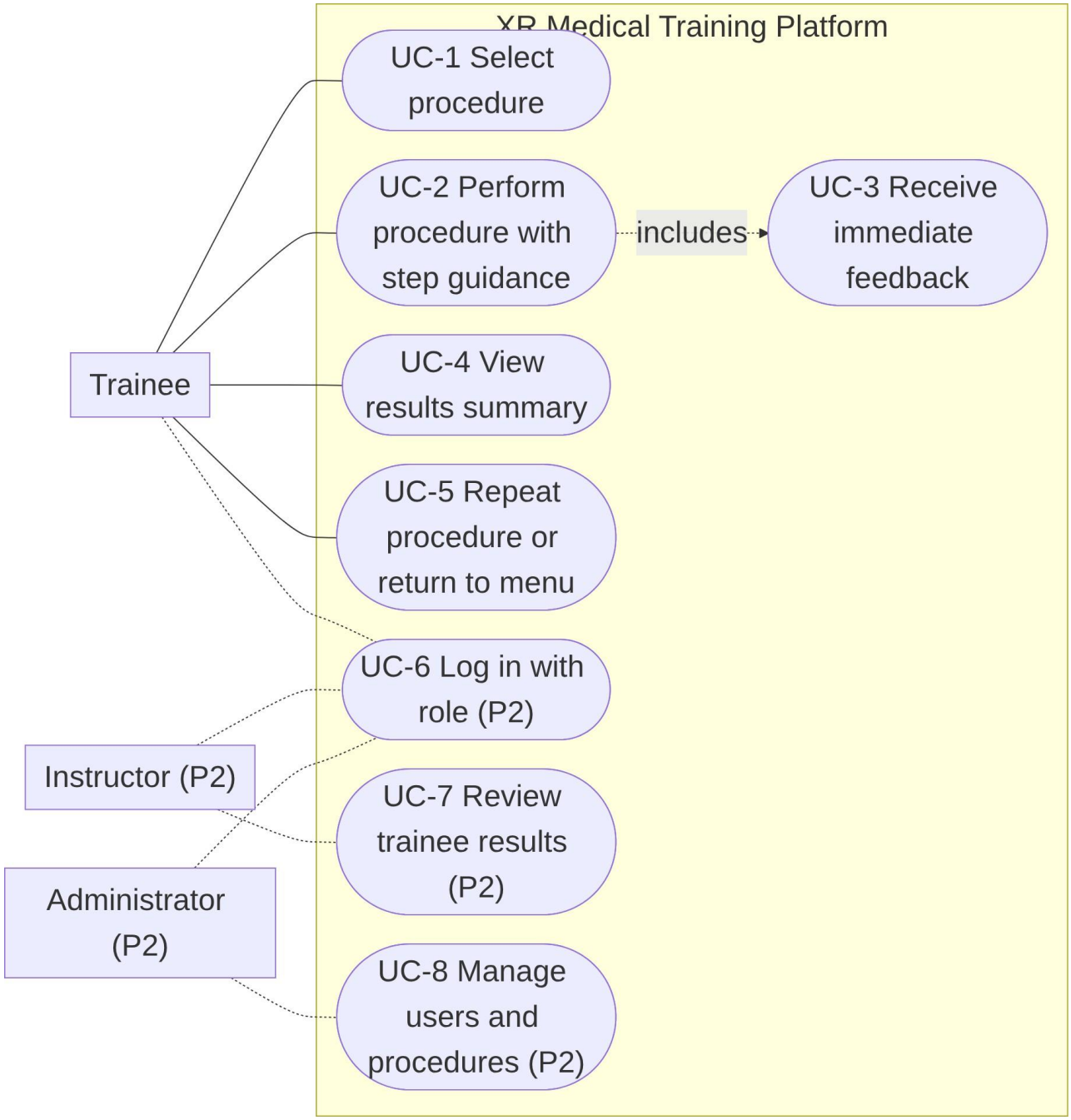


Fig 7.2. Usecase diagram

Use case	Requirements	Stage
UC-1 Select procedure	FR-1	P1
UC-2 Perform procedure with step guidance	FR-2, FR-3, FR-5, FR-6, FR-7, FR-8, FR-9	P1
UC-3 Receive immediate feedback	FR-4	P1
UC-4 View results summary	FR-10	P1
UC-5 Repeat procedure or return to menu	FR-12	P1
UC-6 Log in with role	FR-13	P2
UC-7 Review trainee results	FR-14, FR-15	P2
UC-8 Manage users and procedures	FR-13	P2

Table 7.1 Usecase

7.4 Component diagram (draft)

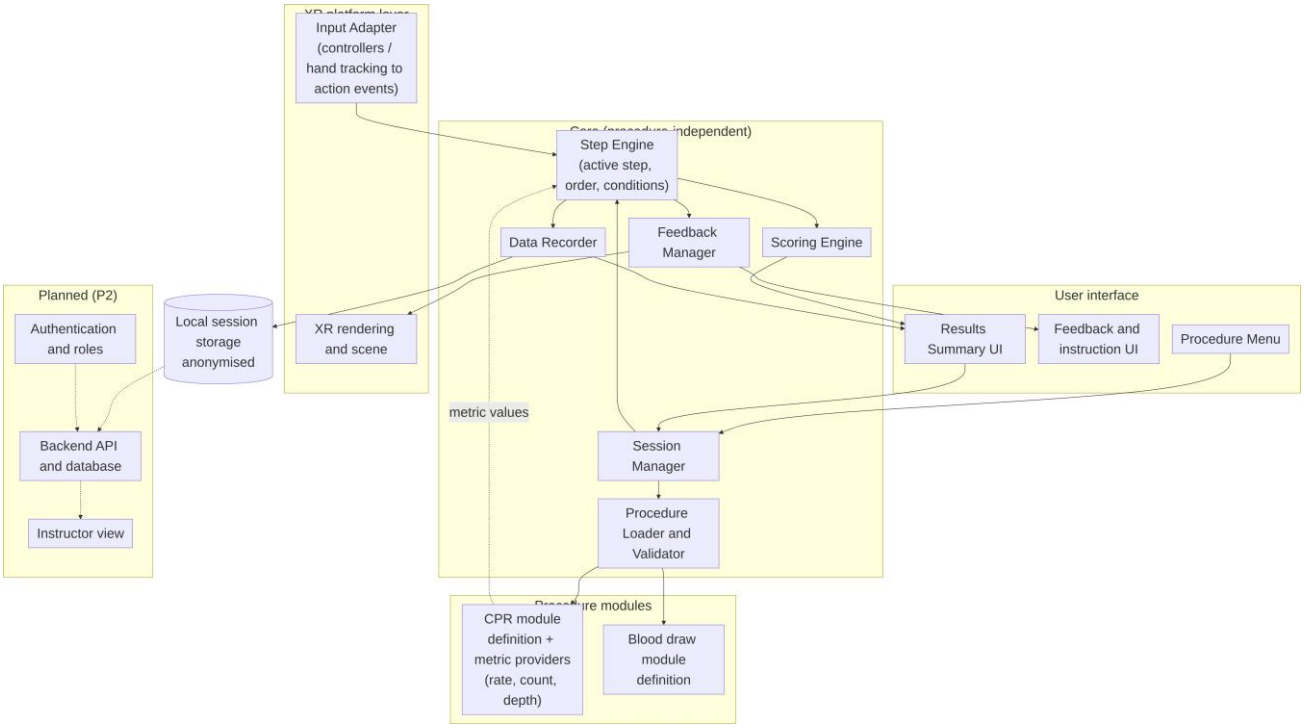


Fig 7.3. Component diagram (draft)

Component	Responsibility	Requirements
Input Adapter	Converts device input into action events; the only place that depends on the input method	FR-5, FR-6, NFR-2
Procedure Menu	Lists available procedures, starts the selected one	FR-1, FR-12
Procedure Loader and Validator	Reads a procedure definition, checks that every step has a condition, a scoring rule and a guideline reference	FR-11, NFR-5, NFR-8
Session Manager	Controls session start, end, restart and return to menu	FR-1, FR-12
Step Engine	Activates steps in order, evaluates conditions, accepts or rejects actions	FR-2, FR-3, FR-7
Metric providers (in procedure modules)	Compute procedure-specific metrics such as CPR rate, count and depth	FR-5, FR-6
Feedback Manager	Shows visual or audio feedback within 1 s after an action	FR-4
Scoring Engine	Applies the scoring rule of each step and computes the session score	FR-8
Data Recorder	Records time and result per step and the CPR metrics; writes anonymised local data	FR-9, NFR-6
Results Summary UI	Shows total time, completed, missed and incorrect steps, score, CPR rate and count	FR-10
Authentication and roles, Backend, Instructor view	Planned for P2	FR-13, FR-14, FR-15

Table 7.2. Components

7.5 Data and procedure definition format

A procedure is described by a definition file (JSON or a Unity ScriptableObject; the final choice is made during implementation). The same format is used for CPR and blood draw (NFR-8).

```
{
  "procedureId": "cpr-basic",
  "name": "Basic CPR",
  "guidelineRef": "<AHA/ERC reference>",
  "steps": [
    {
      "stepId": "S1",
      "order": 1,
      "instruction": "Check responsiveness",
      "condition": { "type": "action", "action": "check_response" },
      "guidelineRef": "<section of guideline>",
      "scoring": { "type": "pass_fail", "points": 1 },
      "feedback": { "correct": "Correct", "incorrect": "Check responsiveness first" }
    }
  ],
  "metrics": [
    { "metricId": "rate", "provider": "CompressionRate", "referenceRange": [100, 120], "unit": "per_min" },
    { "metricId": "count", "provider": "CompressionCount" }
  ]
}
```

Rules of the format:

- Every step has an order, an instruction, a condition, a scoring rule and a guideline reference. Steps without a validated reference are marked `"validated": false` (NFR-5).
- A condition has a type (action, ordered action, metric threshold) and its parameters.
- A scoring rule is at least pass/fail (FR-8).
- Metrics are optional. Their providers are registered by the procedure module, so the core is not changed (FR-5, FR-11).

Session data produced by the Data Recorder:

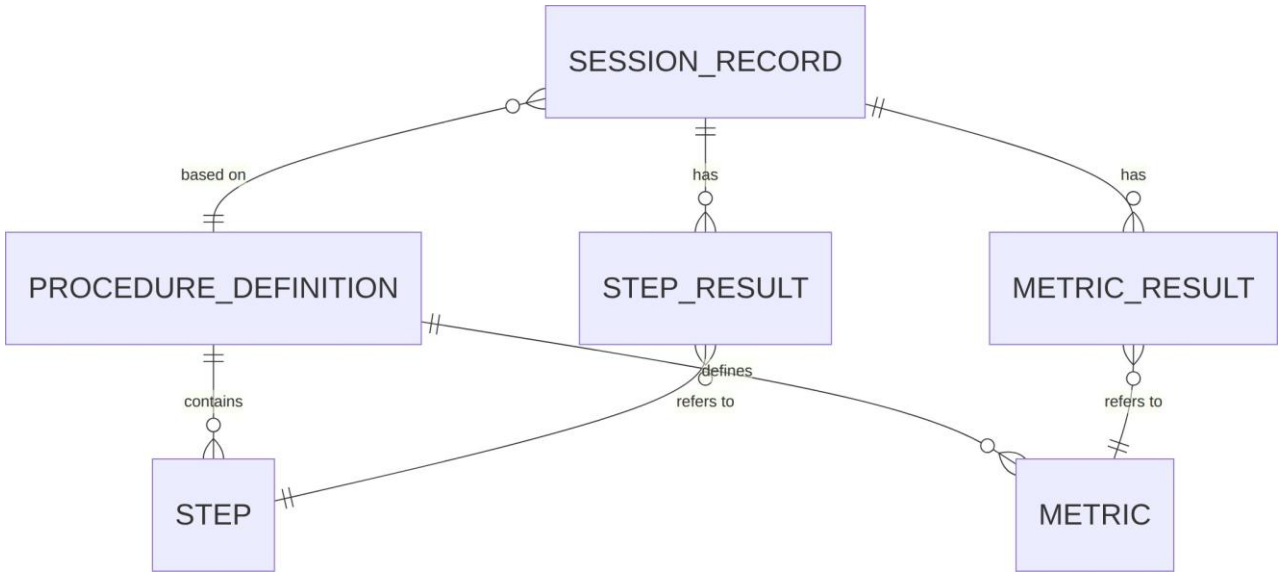


Fig 7.4. Data recorder

Entity	Main fields
SESSION_RECORD	anonymous sessionId, procedureId, start time, end time, total time, score
STEP_RESULT	stepId, time spent, result (passed, missed, incorrect)
METRIC_RESULT	metricId, measured value, within reference range (yes/no)

Table 7.3 Entities

No name, account or device identifier is stored in P1 (NFR-6).

7.6 Main scenario (sequence)

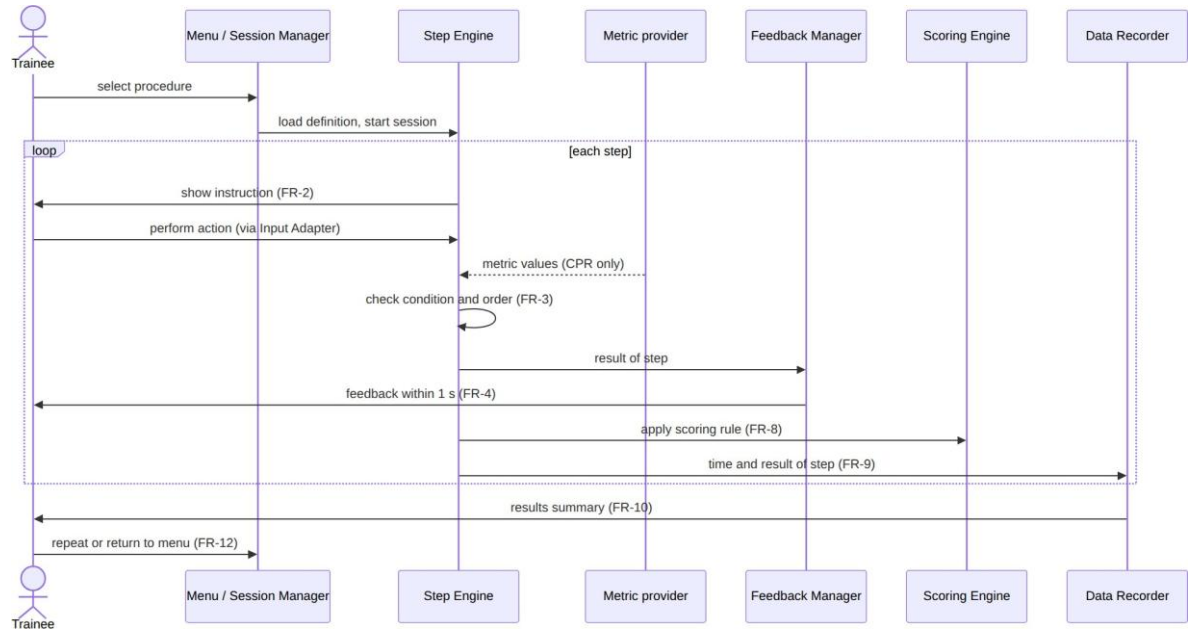


Fig 7.5. Main scenario sequence

Error case: if an action is wrong or out of order, the Step Engine rejects it, the Feedback Manager shows the error, and the step stays active until the trainee corrects it. The result is recorded as incorrect.

7.7 Consistency check: requirements, components, data, scenarios

Requirement	Component	Data	Scenario
FR-1	Procedure Menu, Session Manager	list of procedure definitions	UC-1
FR-2	Step Engine, Feedback and instruction UI	step.instruction	UC-2
FR-3	Step Engine	step.condition, order	UC-2, error case
FR-4	Feedback Manager	step.feedback	UC-3
FR-5, FR-6	CPR metric providers, Input Adapter	metric, METRIC_RESULT	UC-2 (CPR)
FR-7	Step Engine, blood draw module	blood draw definition	UC-2 (blood draw)
FR-8	Scoring Engine	step.scoring	UC-2, UC-4
FR-9	Data Recorder	STEP_RESULT, METRIC_RESULT	UC-2
FR-10	Results Summary UI	SESSION_RECORD	UC-4
FR-11, NFR-8	Procedure Loader and Validator, procedure modules	procedure definition format (7.5)	adding blood draw
FR-12	Session Manager, Menu	none	UC-5
FR-13, FR-14, FR-15	Auth, Backend, Instructor view (P2)	persistent SESSION_RECORD	UC-6, UC-7
NFR-1, NFR-2	XR platform layer, Input Adapter	none	all scenarios on the device
NFR-3, NFR-4, NFR-9	UI, XR rendering	none	UC-2
NFR-5	Procedure Loader and Validator	guidelineRef, validated flag	definition review
NFR-6	Data Recorder, local storage	anonymous SESSION_RECORD	UC-4
NFR-7	whole application	none	10 full sessions

Table 7.4 consistency check

Checks: every Must requirement maps to at least one component, one data item and one scenario; every P1 component serves at least one requirement; P2 components are separated from P1 in all diagrams.

7.8 Open points

- The headset and input method will be fixed in Section 5; the Input Adapter hides the difference, but FR-6 (depth) depends on it.
- The format of the definition file (JSON or ScriptableObject) will be fixed at implementation; the fields in 7.5 stay the same.
- The guideline for blood draw must be selected and listed in Section 12.

8. UX / Prototype

Project: Development of an AR/VR Application for Professional Training of Medical Specialists

This section shows how users interact with the application in the key scenarios of the prototype (P1). The scenarios follow the use cases of Section 7 (UC-1 to UC-5). Each screen is marked as implemented in the Unity prototype or as a mockup.

8.1 Interaction model

- Device: standalone XR headset (Meta Quest), two controllers (Section 5).
- Menus and panels: world-space UI operated with a controller ray and the trigger button.
- Procedure actions: performed in the virtual scene with the controllers (grab tools, press on the manikin chest, and so on); the Input Adapter converts them into device-independent action events (Section 7).
- Session length: up to 15 minutes (NFR-9).
- Design principles, taken from the interviews (Section 2) and the literature (Section 3):
  - guidance for the current step is always visible (FR-2);
  - feedback appears immediately after an action (FR-4, E4);
  - the trainee can repeat a procedure without restarting the application (FR-12, E2);
  - the summary shows mistakes, time and score (FR-10, E7).

8.2 Key user scenarios

Scenario 1: Select a procedure (UC-1)

1. The trainee puts on the headset and sees the main menu with the list of procedures (CPR, blood draw).
2. The trainee points the controller ray at a procedure and presses the trigger.
3. The training scene for the selected procedure loads.

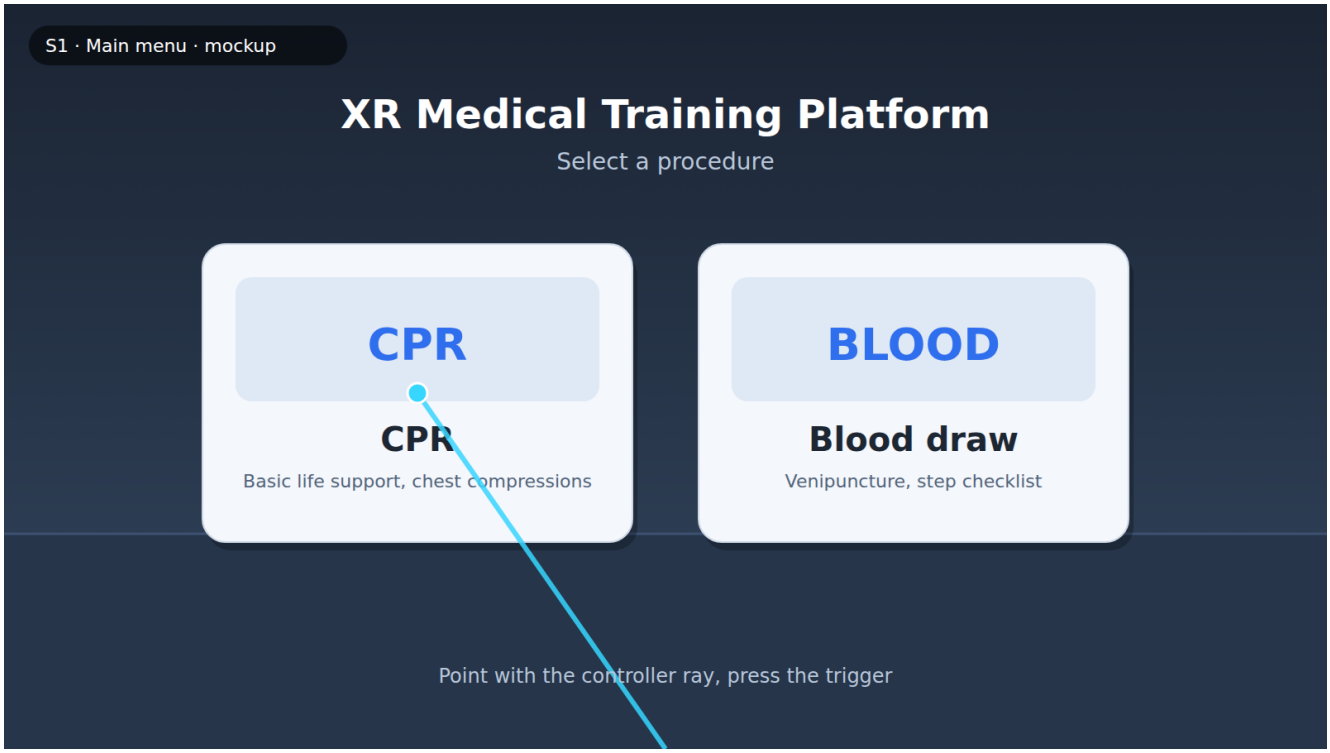


Fig 8.1 S1: main menu (mockup).

Scenario 2: Perform a procedure with step guidance (UC-2, UC-3)

1. The instruction panel shows the current step (for example "Check responsiveness") and the step number.
2. The trainee performs the action in the scene.
3. The system checks the action against the step condition.
4. If the action is correct, the feedback panel shows a confirmation and the next step becomes active.
5. If the action is wrong or out of order, the feedback panel shows the error, and the step stays active until the trainee corrects it.
6. In the CPR procedure, the current compression rate and count are shown during the compression step.

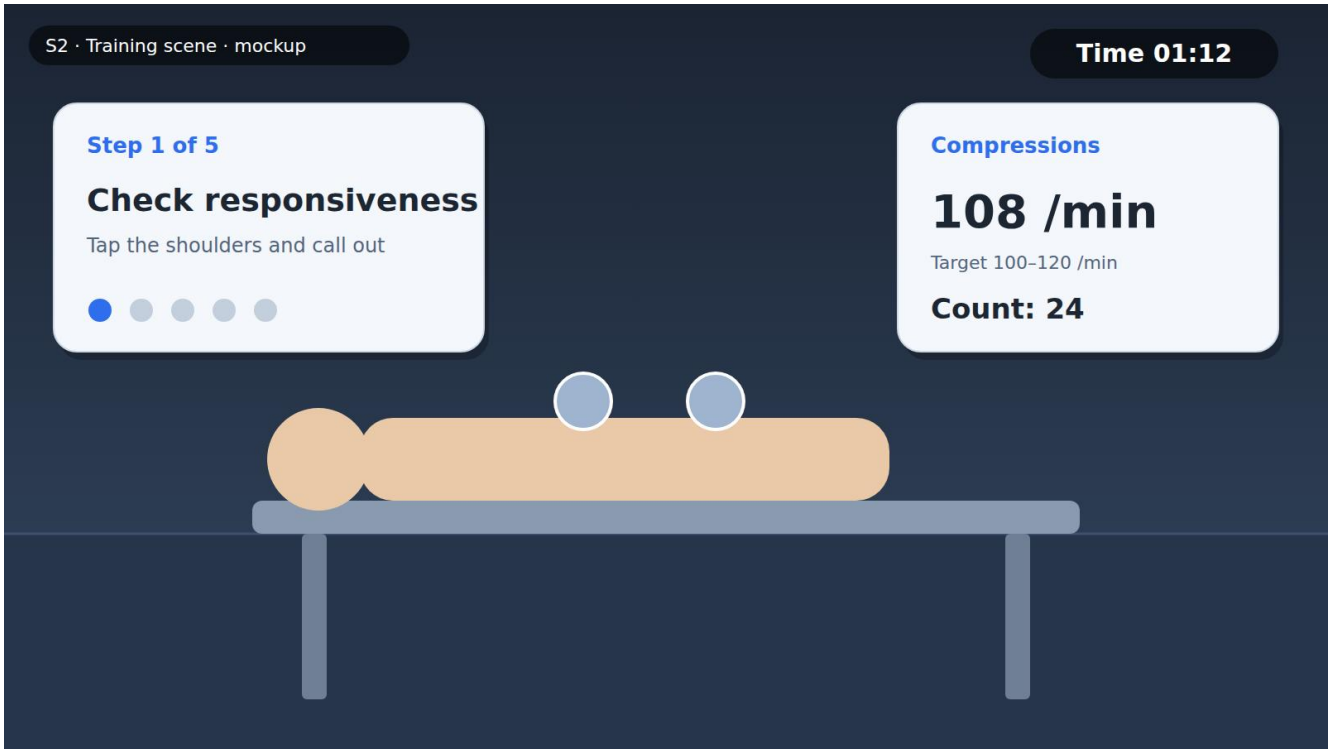


Fig. 8.2. S2: training scene with the instruction panel and CPR metrics (mockup).

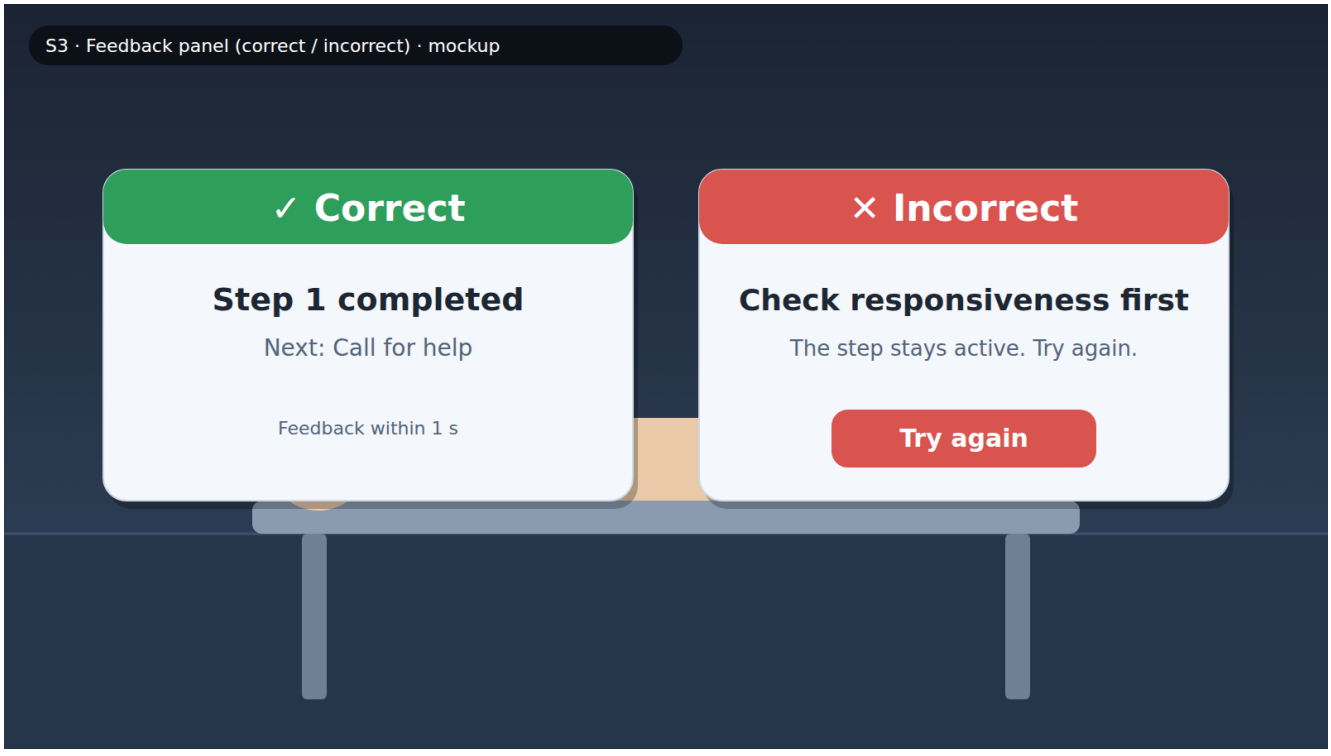


Fig. 8.3. S3: correct and incorrect feedback (mockup).

Scenario 3: View the results summary (UC-4)

After the last step the summary screen shows: total time, number of completed, missed and incorrect steps, score, and for CPR the compression rate and count (FR-10).

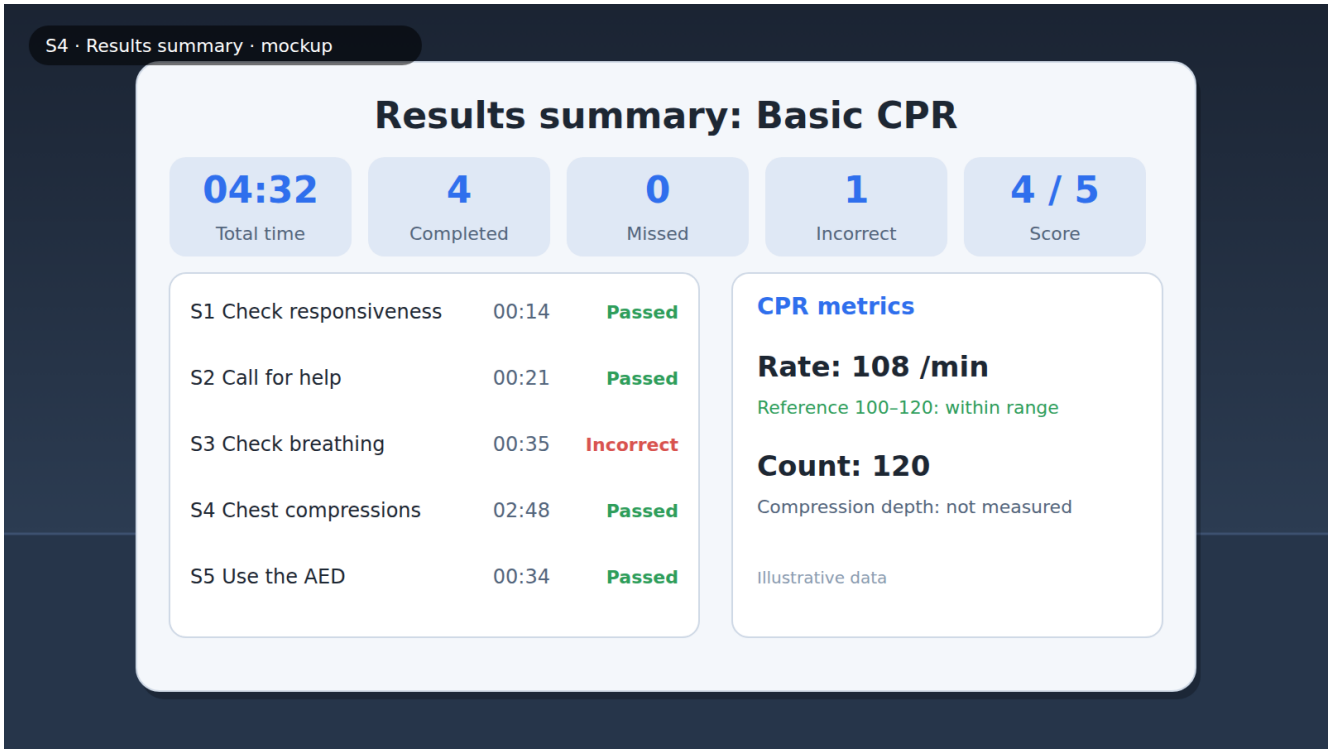


Fig. 8.4. S4: results summary (mockup, illustrative data).

Scenario 4: Repeat or return to the menu (UC-5)

From the summary screen the trainee chooses "Repeat" to restart the same procedure or "Menu" to return to the procedure list (FR-12).

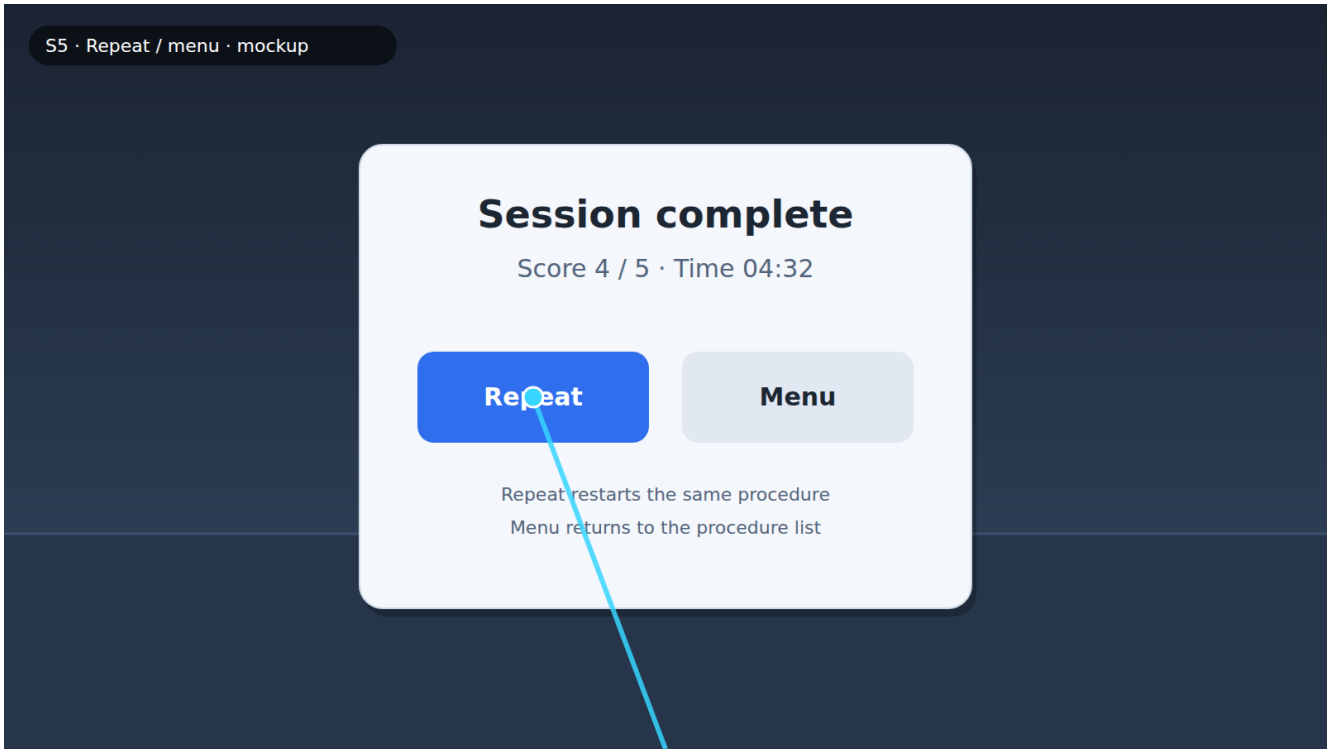


Fig. 8.5. S5: repeat / menu buttons at the end of a session (mockup).

Planned (P2): Instructor view (UC-7)

A mockup of the instructor screen listing trainee sessions with score, time and errors per step. This screen is a mockup only and is not implemented in P1.

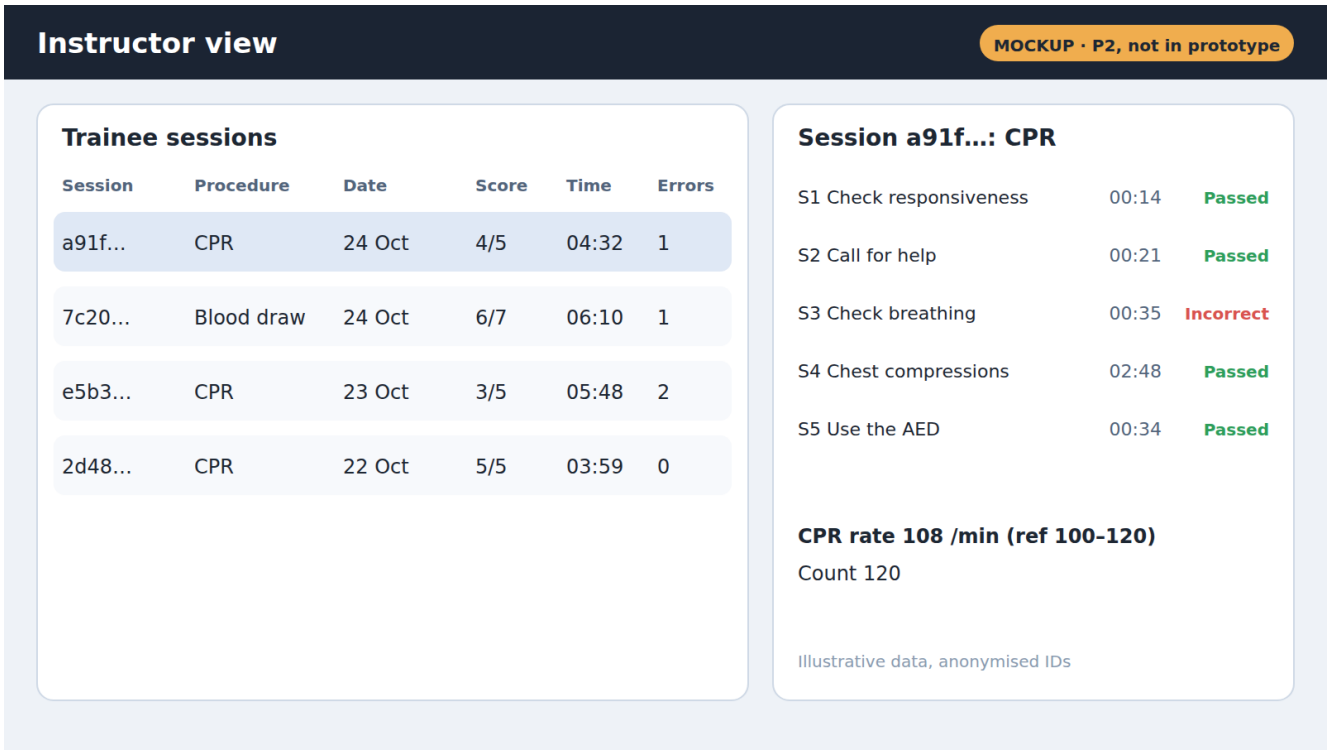


Fig. 8.6. S6: instructor view (mockup, planned for P2, illustrative data).

8.3 CPR procedure flow (activity diagram)

The diagram shows the flow of the CPR procedure from the trainee's point of view, including the error case: a wrong or out-of-order action is rejected, feedback is shown, and the same step stays active until the trainee corrects it (FR-3, FR-4). The step list follows the CPR procedure definition (Section 7.5) and the guideline reference [35]. The sequence of component calls is shown in Section 7.6.

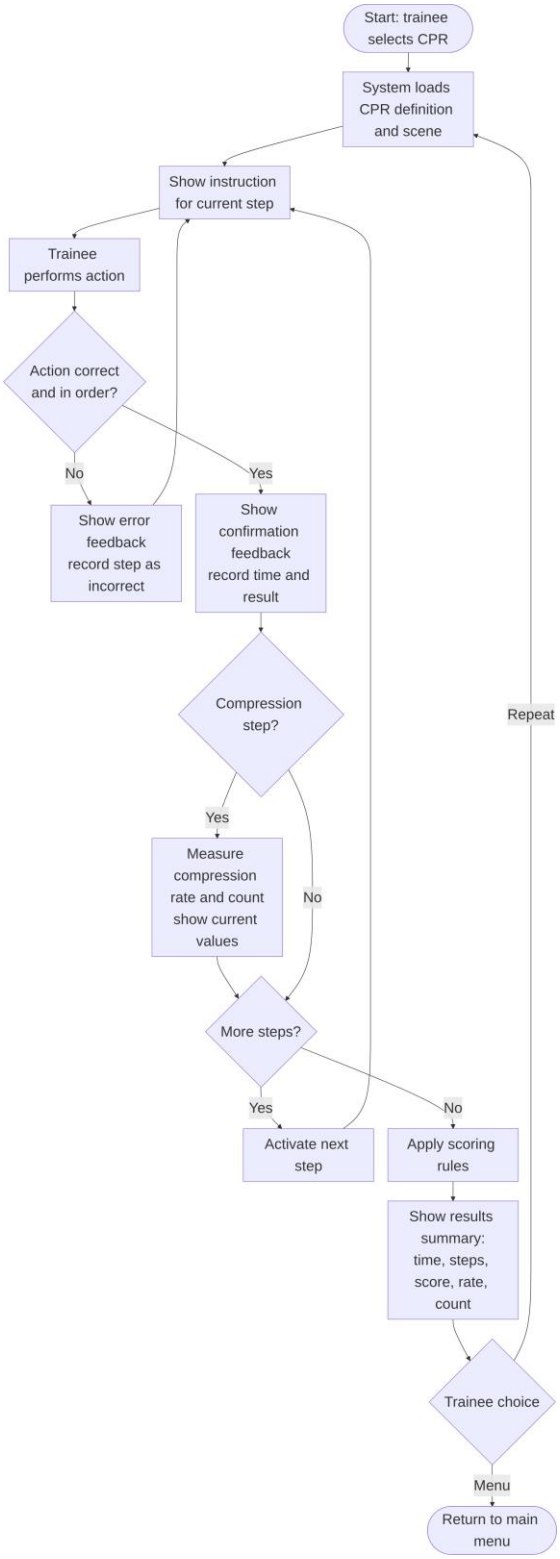


Fig 8.7. CRP procedure flow diagram

Steps of the CPR procedure (to be aligned with the definition file):

Step	Instruction	Condition
S1	Check responsiveness	Action: check_response
S2	<next step, for example: call for help / activate emergency response>	
S3	<next step, for example: check breathing>	
S4	Perform chest compressions	Metric: rate and count within reference range
S5	<optional: use the AED, if included in the prototype>	

8.4 Screen inventory

Screen	Use case	Requirements	Status
S1 Main menu	UC-1	FR-1	<implemented / mockup>
S2 Training scene with instruction panel	UC-2	FR-2, FR-3	<implemented / mockup>
S3 Feedback panel	UC-3	FR-4	<implemented / mockup>
S4 Results summary	UC-4	FR-9, FR-10	<implemented / mockup>
S5 Repeat / menu	UC-5	FR-12	<implemented / mockup>
S6 Instructor view	UC-7	FR-14, FR-15	Mockup, P2

8.5 Interaction table

User action	Input	System response	Requirement
Select a procedure	Controller ray and trigger	Loads the procedure definition and scene	FR-1
Perform a procedure step	Controller action in the scene (grab, press, move)	Checks the condition, shows feedback, activates the next step	FR-2, FR-3, FR-4
Perform chest compressions (CPR)	Controller movement over the manikin chest	Counts compressions and shows the rate; depth is estimated if the input device allows it	FR-5, FR-6
Finish the procedure	Last step completed	Shows the results summary	FR-10
Repeat or return	Controller ray and trigger	Restarts the procedure or opens the menu	FR-12

8.6 Limitations of the prototype

- Screens marked as mockup are not yet implemented in Unity.
- Interaction was tested in the XR Device Simulator; testing on a physical headset depends on getting access to one (Section 9).
- Compression depth depends on the input device and is not yet verified.
- Hand tracking is not used in P1; interaction is controller-based.

9. Verification and Evaluation Plan

This section describes how the achievement of the project aim will be demonstrated: that the application supports hands-on training of CPR and blood draw with step checking, immediate feedback and objective scoring, and that new procedures can be added through a common module format. Thresholds in this plan are proposed targets and will be confirmed with the supervisor.

9.1 Verification levels

Level	Question	Methods	Stage
A. Functional verification	Does each mandatory requirement work as specified?	Acceptance tests with a fixed protocol	Pre-defense 1 (prototype scope), full at pre-defense 2
B. Technical performance	Does the application run smoothly on the target headset?	Unity Profiler, frame time and latency measurement	Pre-defense 1-2
C. Scoring validity	Does automated scoring agree with expert assessment and guideline values?	Comparison against expert checklist and reference values, Cohen's κ	Pre-defense 2

D. Usability and comfort	Is the application usable and tolerable in a 15-minute session?	System Usability Scale (SUS), Simulator Sickness Questionnaire (SSQ)	Pre-defense 2
E. Training effectiveness	Does VR practice improve skill compared with the baseline?	Pilot comparison with a baseline group	Final defense

9.2 Baseline solution

The baseline is conventional training without the VR application: instructional video plus practice on a manikin (CPR) or training arm (blood draw), assessed by an instructor with a checklist. This follows the comparison groups used in the literature: manikin-only simulation in the CPR trial [11], traditional training in the venipuncture trial [5], and video-based modules in the basic care study [9]. The baseline is used in Level E and for reference values in Level C.

9.3 Verification of requirements

Requirement group	Method	Metric	Measurement conditions	Acceptance criterion (proposed)
Procedure execution in VR: CPR and blood draw scenarios run start to finish (FR-1, FR-7)	Scripted acceptance test, each procedure run end to end	Scenario completed without crash or blocking error	Target headset, release build, 10 runs per procedure	10/10 runs complete
Step guidance and step checking (FR-2, FR-3)	Test cases with correct and deliberately wrong actions for each step	Share of steps where the system classifies the action correctly	Fixed test script, same tester, guideline step list as reference	100% of defined steps handled; every wrong action detected in the script
Immediate feedback (FR-4)	Timed test from action to feedback	Feedback delay in seconds	Target headset, 20 actions per procedure	Median delay at most 1 s (the venipuncture system in [5] reported 9 s)
CPR measurement: rate, count, depth (FR-5, FR-6)	Compare system values with reference values from guidelines [35] and with an independent reference measurement or manual count	Rate (compressions/min), count, depth estimate error	Same input device, 5 sessions, fixed duration	Rate and count error at most 5%; depth error limits to be set after checking input-device accuracy [1][6]
Automated scoring and step data (FR-8, FR-9)	Compare automated score with expert checklist score on the same sessions	Cohen's κ (agreement), score difference	At least 10 recorded sessions, scored independently by one expert	κ at least 0.61 (commonly interpreted as substantial agreement); for reference, $\kappa = 0.81$ in [5]
End-of-session review (FR-10)	Acceptance test	Review screen shows per-step result, time and errors	Run after each scenario	Present and correct for all recorded sessions
Extensible procedure module (FR-11, NFR-8)	Add a new test procedure through the module definition without changing core code	Number of core files changed; time to add	One developer, one dummy procedure	0 core files changed
Repeatable practice (FR-12)	Acceptance test	Session can be restarted and repeated	5 repeats in a row	No state carried over incorrectly

Performance (NFR-1)	Unity Profiler and on-device frame measurement	Frame rate, frame time, latency	Target headset, release build, full scenario	Stable at the headset's target refresh rate (72 FPS or higher), no sustained drops; confirm on real device when available
Standalone headset (NFR-2)	Build and run on target device	Runs without PC connection	Meta Quest 3 / 2	Installs and runs standalone
Comfort and session length (NFR-4, NFR-9)	SSQ before and after, session timer	SSQ total score change, symptoms reported, session length	15-minute scenario	Session at most 15 minutes; no participant withdraws because of symptoms
Guideline traceability (NFR-5, R1)	Review of step list against guideline text by an expert	Each step linked to a guideline reference; expert sign-off	Once per procedure	100% of steps referenced; expert comments resolved
Usability	SUS questionnaire	SUS score (0-100)	After the pilot session, each participant	Mean at least 68 (a commonly used benchmark for average usability)

9.4 Evaluation of training effectiveness (Level E)

Goal: show whether VR practice can improve CPR and blood draw skill compared with the baseline, within the limits of a small pilot.

Design:

- Two groups: baseline only versus baseline plus VR practice. The VR group receives the application as an addition to hands-on practice, as in [11], because the literature treats VR as a complement, not a replacement [1][7][10].
- Participants: medical or nursing students on a voluntary basis, with informed consent. Planned sample: as many as can be recruited; the result is a pilot and will not be presented as proof of superiority.
- Procedure: pre-test, training (baseline or baseline plus VR), post-test; optional retention test after 1-2 weeks, because retention is the weakest area of evidence in [1].

Outcome measures:

- Instructor checklist score per step (primary).
- CPR rate and depth versus guideline values, and time to complete the procedure, plus error count.
- Self-reported confidence (short questionnaire).
- SUS and SSQ for the VR

group. Analysis:

- Descriptive statistics and effect sizes; non-parametric tests (for example Mann-Whitney U) because of small samples.
- Limitations are reported openly: small sample, short follow-up, single

institution, possible novelty effect on usability ratings (as noted in [11]).

9.5 Test tooling

- Unity Test Framework (EditMode and PlayMode tests) for step logic, scoring rules and the procedure module loader.
- Unity Profiler and on-device measurement for performance.
- A written manual test protocol (test case ID, linked requirement, steps, expected result, result, date) stored in the Git repository, so that every result is traceable to a requirement and a build version.
- Recorded session data (per-step results, timestamps) exported for the scoring comparison.

9.6 Schedule

Stage	Verification activities
Pre-defense 1	Level A and B for the implemented prototype scope; this plan
Pre-defense 2	Level A complete, Level B on real headset if available, Levels C and D; backend and instructor view tests if implemented
Stage	**Verification activities**
Final defense	Level E pilot and analysis, final acceptance report

9.7 Limitations and risks

- No physical headset is currently available, so on-device performance and comfort measurements (NFR-1, NFR-4) depend on getting access to one. Until then, tests run in the XR Device Simulator and results are marked as simulator-only.
- Depth measurement for CPR depends on the input method; headset and controllers can limit accuracy [1][6]. The accuracy limit will be measured before the depth criterion is fixed.
- Recruiting enough participants for Level E may not be possible; in that case the results are reported as a feasibility study.
- Expert availability for checklist scoring and guideline sign-off is a dependency (R1, R5).

10. Project management and risks

10.1 Git repository

- Repository: https://github.com/tolik-tachyon/xr-medical-training-platform.git
- Structure:
  - `prototype/` — Unity project
  - `docs/` — documentation and pre-defense materials
  - `README.md`

    - `.gitignore` — Unity template
- Every change is linked to a YouTrack task: the task ID is included in the branch name and in the commit or pull request description.
- Branch example: `XMT-16-Project-Management-And-Risks`
- Commit example: `XMT-2: add project proposal content`
- Changes are merged through pull requests; the PR description lists the related YouTrack tasks.

Traceability example

Deliverable	YouTrack task	Branch / commit
Project proposal	XMT-2	commit `XMT-2: add project proposal content`
Project management and risks	XMT-16	branch `XMT-16-Project-Management-And-Risks`
README	XMT-15	commit `XMT-15: prepare README`
Git repository setup	XMT-4	commit `XMT-4: set up repository structure`

10.2 YouTrack project

- Project and board: https://xr-medical-training-platform.youtrack.cloud/agiles/204-1/current
- Board columns: Backlog, Develop, Review, Test, Staging, Done.
- A task is created for each meaningful work item. Each task has an assignee, expected outcome, acceptance criteria, and deadline.
- A link to the relevant commit or pull request is added to the task.
- A task is moved to Done only when its acceptance criteria are met.
- The repository and the board are accessible to the supervisor and the committee.

10.3 Allocation of responsibilities

Role	Name	Responsibilities
Student	Anatoliy Tsoy	All project components: research, requirements, architecture, UX, implementation, verification, documentation
Supervisor	Akerke Akanova	Guidance and review of key deliverables and decisions

10.4 Milestones

Milestone	Date	Content
Internal 1: diagrams ready	23 october	System context, use case and component diagrams (draft)
Internal 2: UX mockups ready	23 october	Mockups of key scenarios: procedure selection menu, CPR, blood draw
Internal 3: supervisor feedback received	23-24 october	Review of draft deliverables, list of comments and decisions
Pre-defense 1	**24 October 2026**	Prototype: a single application with a procedure selection menu and two procedures (CPR and blood draw); proposal, evidence of relevance, literature review, system and technology comparison, requirements, architecture and diagrams, UX mockups, verification plan, project management, supervisor feedback
Pre-defense 2	TBD (to be confirmed by the programme schedule)	User roles, persistent storage of results, instructor view, extended prototype; additional procedures beyond CPR and blood draw are future work
Final defense	TBD (to be confirmed by the programme schedule)	Completed platform, verification results, final report

10.5 Risks

Scales. Probability: Low < 20%, Medium 20–50%, High > 50%. Impact: Medium = delays a deliverable; High = threatens a milestone or the quality of a key deliverable; Critical = threatens the project aim or the safety/correctness of the training content.

Owner of all risks: Anatoliy Tsoy. **Status:** Open unless stated otherwise.

ID	Risk	Prob.	Impact	Mitigation	Contingency (plan B)	Status
R1	Medical inaccuracies in training scenarios may teach incorrect procedures	Medium	Critical	Use reliable medical sources and consult medical professionals to validate procedures and instructions	Limit the prototype to procedures with published step-by-step guidelines; mark unvalidated steps as such	Open
R2	VR performance issues may affect the training experience	High	High	Select the target headset early; develop and test a prototype on the actual device	Reduce model detail and effects; fall back to a simplified scene	Open

R3	Lack of medical knowledge may lead to incorrect procedure modeling	High	High	Limit the prototype to two procedures with published step-by-step guidelines and use validated step-by-step instructions	Ask the supervisor or a medical contact to review the procedure model before it is finalised	Open
R4	Compatibility problems between the headset, Unity version, and XR framework	Medium	High	Fix the Unity and XR package versions early; test a minimal scene on the target device before building features	Switch to the nearest stable Unity/XR package combination	Open
ID	**Risk**	**Prob.**	**Impact**	**Mitigation**	**Contingency (plan B)**	**Status**
R5	Difficulty defining objective scoring rules for the procedures	Medium	High	Use clinical guideline values as reference; check scoring against expert assessment on recorded sessions	Use a simple pass/fail checklist per step instead of a detailed score	Open
R6	Development delays due to limited VR experience and excessive scope	High	High	Prioritize an MVP: complete CPR first, then add blood draw through the procedure module	Reduce blood draw to a pass/fail step checklist without detailed metrics; keep both procedures in the menu (FR-1, FR-7, FR-11)	Open
R7	Limited access to VR equipment or insufficient budget	Medium	High	Confirm equipment availability and costs early; avoid features requiring unavailable hardware	Use the Unity XR device simulator for development only; borrow a headset for acceptance tests; if no device is available, NFR-1 and NFR-2 are reported as not verified in the verification report (Section 9)	Open
R8	Privacy or security issues may arise when storing user data and training results	Low	High	Minimize personal data collection and apply appropriate data protection measures	Store results locally or anonymised until a backend is implemented	Open
R9	Single-person team: illness or overload delays the schedule	Medium	High	Keep tasks small and tracked in YouTrack; plan buffer time before each milestone	Reprioritise to MVP scope; agree deadline adjustments with the supervisor	Open
R10	Limited access to medical experts for interviews and validation	Medium	High	Contact experts early and in parallel with the literature review; prepare short, focused questions	Rely on clinical guidelines and supervisor review; clearly mark findings as unverified	Open

10.6 Dependencies

ID	Task / deliverable	Depends on	Type	YouTrack (task)	YouTrack (depends on)
D1	Requirements specification	Evidence of relevance, Literature review, Existing systems comparison	Finish-to-Start	XMT-8	XMT-3, XMT-5, XMT-6
D2	Architecture and diagrams	Requirements specification, Technology comparison	Finish-to-Start	XMT-9, XMT-10	XMT-8, XMT-7
D3	UX mockups	Requirements specification	Finish-to-Start	XMT-11	XMT-8

D4	Verification and evaluation plan	Requirements specification, metrics, reference values from clinical guidelines (AHA/ERC for CPR, a chosen blood draw guideline)	Finish-to-Start	XMT-12	XMT-8
D5	Prototype (CPR and blood draw)	Technology comparison, Architecture, UX mockups	Finish-to-Start	XMT-17	XMT-7, XMT-9, XMT-11
D6	Second procedure	Architecture and procedure module design, Prototype	Finish-to-Start	XMT-18	XMT-9, XMT-17
D7	Supervisor feedback	Availability of draft deliverables	Finish-to-Start	XMT-13	XMT-8, XMT-9, XMT-11, XMT-12

Link between dependencies and risks: D4 is affected by R5; D5 and D6 by R6; D8 by R1, R3 and R10; D9 by R2, R4 and R7.

The final submission check (XMT-14) depends on all deliverables above.

10.7 Next steps

- Select the target headset and build a minimal scene on it (closes R2, R4, R7).

- Complete the architecture and procedure module design, then add the blood draw procedure through the procedure module without changes to the core (D2, D6, R6).
- Define scoring rules for the procedures based on clinical guideline values (D4, R5).
- Implement a backend with a database to store trainee results (R8).
- Implement user roles and the instructor view so that instructors can review trainee results.
- Schedule validation of the procedures with a medical expert (D8, R1, R3, R10).

11. Supervisor feedback

12. References

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